95.1 1. Free fields and the two principal representatives

Use the free field-strength fields F^a of dim(g) independent Maxwell fields, with the usual physical transverse-photon Fock space. All products in this section are Wick products. In Euclidean notation let

$$J^a_\nu = \sum_{\mu\neq\nu} \partial_\mu F^a_{\mu\nu}, \qquad Q^a_{\nu;\mu} = \partial_\mu F^a_{\mu\nu} - \tfrac{1}{3} J^a_\nu, \qquad H = \sum_{a,\nu} \sum_{\mu\neq\nu} :(Q^a_{\nu;\mu})^2: . \tag{DN1}$$

Define the cubic representative by

$$C = \sum_{a,b,c} \sum_{\mu,\nu,\rho} f_{abc} :F^a_{\mu\nu} F^b_{\nu\rho} F^c_{\rho\mu}:, \tag{DN2}$$

up to a fixed nonzero normalization of the invariant structure tensor. Any compact simple Lie algebra has a triple of generators for which f_{abc} is nonzero. C is the principal continuum cubic in AI's exact lattice C-sharp. H is the principal directional-current quadratic after the full current has been completed. EOM terms in the free theory are contact distributions; they do not affect the positive-time norms below.

Let f be a real smooth test function with support in t>=t0>0. Write

$$\widetilde{f}(E,\mathbf{k}) = \int_0^\infty dt \int d^3x \; f(t,\mathbf{x}) e^{-Et + i\mathbf{k}\cdot\mathbf{x}}. \tag{DN3}$$

The free reflected norm of a Wick monomial of field degree n is a positive integral of the form

$$\|[O(f)]\|_{\mathrm{OS}}^2 = c_n \sum_{\text{colors, polarizations}} \int \prod_{i=1}^{n} \frac{d^3\mathbf{p}_i}{2|\mathbf{p}_i|} \left| \widetilde{f}\left(\sum_i |\mathbf{p}_i|, \sum_i \mathbf{p}_i\right)\right|^2 |\mathcal{M}_O(p_1,\dots,p_n)|^2, \qquad c_n > 0. \tag{DN4}$$

This follows directly by expanding the positive-energy creation part of each free field and performing the time and space smearing. Wick ordering leaves exactly n creation operators on the vacuum; their Fock norm gives the polarization sum and positive phase-space measure. Fixed Fourier conventions only change c_n. The support t>=t0 supplies exponential decay in the sum of energies, so the integral is finite for the fixed polynomial form factors used here. Field-strength factors also give integrability at zero momentum. No extension across the reflected coincidence diagonal is needed because t+t'>=2t0.

Thus strict positivity reduces to a nonzero on-shell form factor on an open set. We check that property explicitly for both directions.

95.2 2. The quadratic direction survives the free field equations

Use complex Euclidean on-shell momentum and polarization

$$p = (i, 1, 0, 0), \qquad e = (0, 0, 1, 0), \qquad F_{\mu\nu} = p_\mu e_\nu - p_\nu e_\mu. \tag{DN5}$$

Then p squared and p dot e vanish. Therefore

$$\sum_\mu p_\mu F_{\mu\nu} = 0, \qquad \sum_{\mu,\nu} (p_\mu F_{\mu\nu})^2 = 2. \tag{DN6}$$

Indeed the only nonzero directional-current terms are -1 and +1 for nu=2 and mu=0,1. Their sum is zero and their squares sum to two. The J/3 terms in DN1 vanish on shell. The two-photon form factor with equal colors and these equal momenta is consequently nonzero; symmetrizing the two legs multiplies it by a nonzero fixed factor. Equal momenta are only used to find one point. Continuity makes it nonzero on a neighborhood of physical transverse momenta, which has positive phase-space measure.

This is a physical on-shell test. Nonzero norm of an unconstrained Euclidean polynomial alone would not exclude an equation-of-motion contact field.

95.3 3. The cubic direction also has a nonzero physical form factor

For a completely explicit check, use Minkowski metric eta=diag(1,-1,-1,-1) and the future null vectors and transverse polarizations

$$\begin{aligned} p_1 &= (1,1,0,0), \quad e_1 = (0,0,1,0), \\ p_2 &= (1,0,1,0), \quad e_2 = (0,0,0,1), \\ p_3 &= (1,0,0,1), \quad e_3 = (0,1,0,0). \end{aligned} \tag{DN7}$$

Let (F_i)$^{\text{mu_nu=p_i}}$mu(e_i)_nu-e_i^mu(p_i)_nu. Each p_i has squared norm zero and p_i dot e_i=0. Multiplying the three four-by-four matrices gives

$$\mathrm{tr}_{\mathrm{Lor}}(F_1F_2F_3) = -1. \tag{DN8}$$

For example, in this product the nonzero contribution to the trace is the spatial cycle 1 to 2 to 3 to 1, with three entries -1. For eta-skew matrices, transpose followed by eta conjugation shows tr(XYZ)=-tr(ZYX). Together with cyclicity this makes tr(XYZ) an alternating trilinear form. Choose three colors with f_{123} nonzero. When all assignments of these three external legs to DN2 are summed, the sign of the structure tensor and the sign of the Lorentz trace change together. The contributions add, rather than cancel. The cubic form factor is a nonzero multiple of 6 f_{123} times DN8.

Again continuity gives a nonzero form factor on an open set of physical momenta and polarizations. Analytic continuation of the time indices yields the Euclidean reflected state in DN4, with the same nonvanishing conclusion.

95.4 4. A positive two-by-two free Gram matrix

Choose one test f for which DN3 is nonzero near both momentum sums just used. Such a smooth compactly supported test can be chosen explicitly as a product of a nonnegative time bump in (t0,2t0) and a nonnegative spatial bump of sufficiently small support. Its time Laplace transform is strictly positive. Its spatial Fourier transform is nonzero near either of the two fixed sums when the spatial support is sufficiently small.

The state H(f) has exactly two free particles and C(f) has exactly three. Fock sectors of different particle number are orthogonal. Equations DN4–8 therefore give

$$G_0 = \begin{pmatrix} \langle H(\theta f)H(f)\rangle_0 & \langle H(\theta f)C(f)\rangle_0 \\ \langle C(\theta f)H(f)\rangle_0 & \langle C(\theta f)C(f)\rangle_0 \end{pmatrix} = \begin{pmatrix} h_0 & 0 \\ 0 & c_0 \end{pmatrix}, \qquad h_0, c_0 > 0. \tag{DN9}$$

In these expressions theta includes the usual reflection/conjugation of tensor indices. Both complete scalar contractions are even under time reflection. Let m0=min(h0,c0).

For any Hermitian physical Gram matrix G satisfying

$$\|G - G_0\|_{\mathrm{op}} < \tfrac{1}{2} m_0, \tag{DN10}$$

one has G>=m0 I/2. Its inverse and positive inverse square root are bounded, and a change of basis by that inverse square root preserves reflection positivity. This proves that the coupled normalization problem has no degeneracy at the free reference. It does not prove DN10 for the actual interacting lattice insertions.

95.5 5. Noncommuting matrix mixing has an elementary stable normalization

Here is the precise finite-dimensional transport lemma. Suppose a renormalization step on a fixed d-dimensional operator quotient has the actual transfer matrix

$$T_j = L^{-\gamma}(I + E_j), \qquad \|E_j\| \le C/x_j \le \tfrac{1}{2}, \tag{DN11}$$

where the inverse squared couplings decrease and b_-<=x_j-x_{j+1}<=b_+ with b_->0. Put

$$P_{m:n} = (I + E_{n-1}) \cdots (I + E_m), \qquad m < n. \tag{DN12}$$

Every factor is invertible. Multiplying the Neumann bounds gives

$$\|P_{m:n}\| + \|P_{m:n}^{-1}\| \le 2\exp\left(2C\sum_{j=m}^{n-1} x_j^{-1}\right) \le 2(x_m/x_n)^{2C/b_-}. \tag{DN13}$$

For the last inequality note that integral from x_{j+1} to x_j of dx/x is at least (x_j-x_{j+1})/x_j>=b_-/x_j, and sum. No commutativity, diagonalizability, or absence of resonances is assumed. The ordered inverse product is the normalization; diagonalizing each step independently is unnecessary.

If the actual estimate is only $\|E_j\| \le C/\sqrt{x_j}$, the same proof gives

$$\|P_{m:n}\| + \|P_{m:n}^{-1}\| \le 2\exp\{(4C/b_-)(\sqrt{x_m} - \sqrt{x_n})\}. \tag{DN14}$$

This weaker bound is stated separately: an O(g) interaction must not be silently replaced by an O(g squared) estimate just to obtain a power of the logarithm. For the present dimension-six candidate the engineering value would be gamma=2 in four dimensions, after the actual local projection and lower-dimensional subtractions have been specified. This chapter does not assert DN11 for that projection.

Pointwise monotonicity is not necessary for the useful bound. Suppose instead the actual trajectory has the lower envelope

$$x_j \ge x_n + c(n-j) - C_0 \quad (m \le j < n), \qquad x_n > C_0, \qquad c > 0. \tag{DN15}$$

Put y=x_n-C0. Comparing the decreasing function 1/(y+ct) with its integral on [0,n-m] gives

$$\sum_{j=m}^{n-1} x_j^{-1} \le \frac{1}{c}\log\frac{y + c(n-m)}{y} \le \frac{1}{c}\log\frac{x_m}{x_n - C_0}. \tag{DN16}$$

Consequently the first Neumann-product bound in DN13 yields 2(x_m/(x_n-C0))^(2C/c) under DN15 as well. This is the form appropriate to a trajectory controlled by an affine lower envelope and bounded partial-sum errors. It still requires the actual matrix estimate in DN11; no trajectory estimate proves that operator estimate.

95.6 6. The remaining interacting statement

To turn this normalization into a theorem for the Wilson measure, one still has to construct its exact completed two-component insertion activity, including the lower-dimensional mixings and contact terms, and prove its transfer estimate and its remainder estimate on the common marked domains. Then one needs a separated physical comparison strong enough to imply DN10. Neither step follows from nondegeneracy of G0.

The new CL protected hull controls the first-derivative curvature expressions occurring in H and the cubic expression occurring in C-sharp. It is therefore relevant to this two-component problem. It does not bound arbitrary higher spatial jets, as HJ shows. For higher dimensions the corresponding finite quotient, its completion, and its Wilson-measure estimates still have to be constructed. The free calculation here removes a specific possible normalization obstruction; it does not close that hierarchy.

96.1 1. The exact input and the path conventions

Work in a fixed physical four-dimensional box with a collar, or on a torus. Let the microscopic spacing be a, and write

$$(X,Y)_a = a^4 \sum_x \langle X(x), Y(x)\rangle, \qquad \langle X,Y\rangle = -\operatorname{tr}(XY). \tag{CD1}$$

The pairing is symmetric and invariant under the complexified gauge group. Bounds use absolute matrix norms; in particular the pairing in (CD1) is not assumed positive for complex fields.

On a controlled local analysis chart let $U_i(x) = \exp(aA_i(x))$, with

$$\|A\|_{C^{1,\beta}} \le M, \qquad \|F_a\|_{C^\beta} \le M_F, \qquad 0 < s < \tfrac{1}{2}, \quad s \le \beta < 1. \tag{CD2}$$

Here $F_{a,ij}(x) = a^{-2}\log U_{ij}(x)$ is the actual based logarithmic plaquette curvature, on its fixed small branch. A finite change to a different analytic compact curvature completion can be accounted for separately; it is not made inside the identities below. The lattice Hölder and Sobolev norms are the ordinary physical-mesh norms in this chart. Local norms are taken on a larger collar containing all translated supports and paths.

The CL/MC input implies (CD2), uniformly in a, with $p > 8$, $0 < s < 1/p$, and a choice

$$s < \beta < 1 + s - \frac{4}{p}. \tag{CD3}$$

Indeed its total-field Coulomb chart has $A \in W^{2+s,p}$, while its curvature is in $W^{1+s,p}$. The present theorem needs only the displayed lower norms and $X, Y \in H^{1+s}$, when X, Y are selected curvature components. It does not demand pointwise second derivatives of curvature.

For an oriented nearest-neighbor step e put

$$T_e X(x) = \operatorname{Ad} U(x, x+e)\, X(x+e), \qquad D_i^U X = a^{-1}(T_{ae_i} - I)X. \tag{CD4}$$

The negative step uses the actual inverse link, so $T_{-ae_i} = T_{ae_i}^{-1}$. The forward covariant derivative is fixed throughout; reverse steps are explicitly shifted to this convention.

Let a path $\gamma = (v_1, \dots, v_N)$, $v_j = \pm a e_i$, have displacement $h = \sum_j v_j$ and length $\ell = Na$. Its translated parallel transport defines T_γ. Write

$$\bar\gamma = (-v_1, \dots, -v_N), \qquad \gamma^{-1} = (-v_N, \dots, -v_1). \tag{CD5}$$

Thus $\bar{\gamma}$ is the spatially reflected path and γ^{-1} is the reversed path. Both have displacement $-h$, but they are different paths in a nonflat field.

For a scalar test f, sampled also at the geometric midpoints, define

$$\begin{aligned} B_\gamma(f;X,Y) &= a^4 \sum_x f(x+h/2)\langle X(x), T_\gamma Y(x)\rangle, \\ \mathcal{B}_\gamma(f;X,Y) &= \tfrac{1}{2}\{B_\gamma(f;X,Y) + B_{\bar{\gamma}}(f;X,Y)\}. \end{aligned} \tag{CD6}$$

The reflected term has midpoint $x - h/2$. Midpoints need not be lattice sites: only the smooth scalar test is evaluated there.

The signed area between γ and its straight chord is

$$\mathcal{A}_{ij}(\gamma) = \frac{1}{2}\sum_{k<l}(v_k^i v_l^j - v_k^j v_l^i), \qquad i<j. \tag{CD7}$$

This is a geometric number, with $\sum_{i<j}|\mathcal{A}_{ij}| \le C\ell^2$. No continuum interpolation of the connection occurs in this definition. The orientation of $F_{a,ij}$ is that of the word $+i,+j,-i,-j$, so the signs in the area term below are fixed.

96.2 2. A cutoff-uniform covariant first-difference lemma

For $0<s<1$, $X \in H^{1+s}$, and paths of length $\ell \le 1$,

$$\begin{aligned} \|T_\gamma X - X\|_2 &\le C_M \ell \|X\|_{H^1}, \\ \left\| T_\gamma X - X - \sum_i h_i D_i^U X \right\|_2 &\le C_{M,s}\ell^{1+s}\|X\|_{H^{1+s}}. \end{aligned} \tag{CD8}$$

These estimates include paths with reverse steps and closed paths. The second estimate uses the net displacement h, not the unsigned number of steps in each direction.

Here is a direct proof. First,

$$D_i^U X = D_i^+ X + B_i(x)X(x+ae_i), \qquad B_i = a^{-1}(\operatorname{Ad} e^{aA_i} - I). \tag{CD9}$$

The analytic formula for B_i gives a uniform C^1 bound on the fixed A ball, so multiplication by B_i preserves H^s. Consequently $\|D_i^U X\|_{H^s} \le C_M\|X\|_{H^{1+s}}$.

A translated path of length $d \le 1$ has $\|\operatorname{Ad} P - I\|_\infty \le C_M d$. Combining this with the ordinary discrete fractional translation inequality gives

$$\|(T_\pi - I)G\|_2 \le C_{M,s} d^s \|G\|_{H^s}. \tag{CD10}$$

The constants do not count microscopic path bonds. For a negative step,

$$T_{-ae_i} - I = -aT_{-ae_i}D_i^U. \tag{CD11}$$

Telescope the full word for $T_\gamma - I$; in each summand use (CD11) when required and replace the transported $D_i^U X$ by its value at the root. Each replacement costs at most $Ca\ell^s\|X\|_{H^{1+s}}$, and there are $N = \ell/a$ steps. Their constant parts sum to $\sum_i h_i D_i^U X$. This proves (CD8). The first inequality is the same telescope with $s=0$.

All of this remains valid for complex A. Along a path of bounded physical length its transports have uniformly bounded norms and inverse norms. The trace pairing remains exactly invariant.

96.3 3. The inverse-paired quadratic identity and its strict remainder

Set

$$\mathcal{I}_\gamma = \tfrac{1}{2}\{B_\gamma + B_{\gamma^{-1}}\}, \qquad \delta_\gamma = T_\gamma - I.$$

Changing a lattice summation variable by h, and using invariance of the pairing, gives the EXACT identity

$$\begin{aligned}\mathcal{I}_\gamma(f;X,Y) = \frac{1}{2}a^4\sum_x\{f(x+h/2)+f(x-h/2)\}\langle X,Y\rangle(x)\\ -\frac{1}{2}a^4\sum_x f(x+h/2)\langle\delta_\gamma X,\delta_\gamma Y\rangle(x).\end{aligned} \tag{CD12}$$

In particular this is a bilinear identity for different curvature components. It does not require $X = Y$, positivity, or a square root of the test.

Taylor expansion of the scalar test and (CD8) now imply

$$\begin{aligned}\mathcal{I}_\gamma(f;X,Y) &= (fX,Y)_a - \frac{1}{2}\sum_{i,j}h_ih_j(fD_i^UX,D_j^UY)_a\\ &\quad+\frac{1}{8}\sum_{i,j}h_ih_j((\partial_i\partial_jf)X,Y)_a + R_{\mathrm{inv}},\\ |R_{\mathrm{inv}}| &\le C_{M,s}\ell^{2+s}\|f\|_{C^{2,s}}\|X\|_{H^{1+s}}\|Y\|_{H^{1+s}}.\end{aligned} \tag{CD13}$$

To check all powers, the product of the leading first differences has size $O(\ell^2)$. Its product with one remainder from (CD8) costs ℓ^{2+s}, and two remainders cost ℓ^{2+2s}. Replacing $f(x+h/2)$ by $f(x)$ in that leading product costs $O(\ell^3)$, which is better because $s<1$. The central Taylor remainder of the smooth test is $O(|h|^{2+s})$. These are the only replacements in (CD13).

If literal lattice descendants are preferred, replace $\partial_i\partial_jf$ by any fixed consistent second lattice difference of f. Their difference is bounded by $Ca^s\|f\|_{C^{2,s}}$, and $a\le\ell$, so the resulting change is again covered by (CD13). Summation by parts then makes it an exact descendant of the chosen quadratic completion.

96.4 4. Reflection is not reversal: the local cubic area term

The distinction in (CD5) contributes at dimension six. The relative holonomy

$$K_\gamma(x) = P_{\bar\gamma}(x)P_{\gamma^{-1}}(x)^{-1} \tag{CD14}$$

is the transport round the closed path consisting of $\bar\gamma$ and the translated original γ. Its signed area is exactly $2\mathcal{A}_{ij}(\gamma)$.

For $\ell\le 1$, on the small core (CD2), an exact plaquette-word reduction gives

$$\begin{aligned}K_\gamma(x) &= I + 2\sum_{i<j}\mathcal{A}_{ij}(\gamma)F_{a,ij}(x) + E_\gamma(x),\\ \|E_\gamma\|_\infty &\le C_{M,M_F}\ell^{2+\beta}.\end{aligned} \tag{CD15}$$

The same estimate holds for its adjoint action, with the linear term replaced by the corresponding commutator.

For clarity, this estimate is a statement about actual plaquettes. Reorder the closed nearest-neighbor word by adjacent swaps and cancel oppositely oriented equal-axis steps. At most CN^2 swaps are

needed. Each swap inserts a plaquette or its inverse conjugated by a path of length $C\ell$. The signed plaquette count is the discrete signed area of the original word. Each plaquette is exactly $\exp(a^2 F_{a,ij})$ at its own base. Transporting that curvature to x and replacing its argument by x costs $C\ell^\beta$ in norm. Multiplying the sum of at most CN^2 such errors by a^2 gives $C\ell^{2+\beta}$. The linear transport-conjugation error $C\ell M M_F$ contributes $C\ell^3$; the product remainder of the small plaquette factors is bounded by $C(\ell^2 M_F)^2 \exp(C\ell^2 M_F)$. Both fit (CD15). This proof uses only the admitted C^β curvature and bounded path transports, not a pointwise derivative of curvature.

Since $T_{\bar\gamma} Y = \operatorname{Ad} K_\gamma \, T_{\gamma^{-1}} Y$, (CD15), (CD8), and a first Taylor estimate for f give

$$\begin{aligned} \tfrac{1}{2}(B_{\bar\gamma} - B_{\gamma^{-1}}) &= \sum_{i<j} \mathcal{A}_{ij}(\gamma)(fX, [F_{a,ij}, Y])_a + R_{\text{area}}, \\ |R_{\text{area}}| &\le C_{M,M_F} \ell^{2+\beta} \|f\|_{C^1} \|X\|_{H^1} \|Y\|_{H^1}. \end{aligned} \tag{CD16}$$

Replacing $T_{\gamma^{-1}} Y$ by Y in the leading term costs $C\ell^3 \|X\|_2 \|Y\|_{H^1}$, and replacing the shifted test costs the same order. These replacements are explicitly included.

Combining (CD13) and (CD16) proves the main theorem:

$$\boxed{\begin{aligned} \mathcal{B}_\gamma(f; X, Y) &= (fX, Y)_a - \frac{1}{2} h_i h_j (f D_i^U X, D_j^U Y)_a \\ &\quad + \frac{1}{8} h_i h_j ((\partial_i \partial_j f) X, Y)_a \\ &\quad + \sum_{i<j} \mathcal{A}_{ij}(\gamma)(fX, [F_{a,ij}, Y])_a + R_\gamma, \\ |R_\gamma| &\le C_{M,M_F,s,\beta} \ell^{2+s} \|f\|_{C^{2,s}} \|X\|_{H^{1+s}} \|Y\|_{H^{1+s}}, \qquad s \le \beta. \end{aligned}} \tag{CD17}$$

Thus the complete second engineering-order projection contains the full derivative-curvature quadratic tensor, the full quadratic descendant, and the path-area curvature commutator. Symmetry can reduce this list only after the full pointed carrier and its reflected partners have been combined. In particular an asymmetric individual carrier is not declared even in isolation.

For $X = Y$, the displayed commutator pairing vanishes pointwise. For different curvature components it need not vanish. Averaging independently in the field labels X, Y would erase that antisymmetric color structure and is not a legitimate substitute for the physical path reflection in (CD6).

The area commutator is charge-conjugation even for curvature fields. It belongs to the even curvature-cubic sector, not to the protected 56-dimensional C-odd cubic center.

96.4.1 4.1 A pointed source need not sit at the midpoint

There is an explicit repointing formula, so a source anchor is not lost when a carrier is put in midpoint form. Define $B_{\gamma,z}$ by replacing $f(x + h/2)$ in (CD6) with $f(x + z)$, and pair it with $B_{\bar\gamma,-z}$. Assume $|z| \le C\ell$, put $d = z - h/2$, and write $D_h^U = \sum_i h_i D_i^U$, $\partial_z = \sum_i z_i \partial_i$. Then

$$\begin{aligned} \tfrac{1}{2}(B_{\gamma,z} + B_{\bar\gamma,-z}) &= (fX, Y)_a - \tfrac{1}{2}(f D_h^U X, D_h^U Y)_a \\ &\quad + ((\partial_d f) X, D_h^U Y)_a + \tfrac{1}{2}((\partial_z^2 f) X, Y)_a \\ &\quad + \sum_{i<j} \mathcal{A}_{ij}(\gamma)(fX, [F_{a,ij}, Y])_a + R_{\gamma,z}, \end{aligned} \tag{CD17a}$$

with the same $C\ell^{2+s}$ bound as (CD17).

To verify this without a second field derivative, set

$$f_{\mathrm{e}} = \tfrac{1}{2}\{f(\cdot + d) + f(\cdot - d)\}, \qquad f_{\mathrm{o}} = \tfrac{1}{2}\{f(\cdot + d) - f(\cdot - d)\}.$$

The pointed expression is exactly the midpoint-even expression with test f_{e}, plus the midpoint-odd expression with test f_{o}. Taylor gives $f_{\mathrm{e}} = f + \frac{1}{2}\partial_d^2 f + O(\ell^{2+s})$ and $f_{\mathrm{o}} = \partial_d f + O(\ell^{1+s})$. Applying (CD8) to the odd expression gives

$$\tfrac{1}{2}(B_\gamma(f_{\mathrm{o}}) - B_{\bar\gamma}(f_{\mathrm{o}})) = (f_{\mathrm{o}}X, D_h^U Y)_a + \tfrac{1}{2}((\partial_h f_{\mathrm{o}})X, Y)_a + O(\ell^{2+s}). \tag{CD17b}$$

Its first-difference remainder has one additional factor $\|f_{\mathrm{o}}\|_{C^{1,s}} \le C\ell\|f\|_{C^{2,s}}$. Collecting the scalar second derivatives gives $\frac{1}{8}\partial_h^2 + \frac{1}{2}\partial_h\partial_d + \frac{1}{2}\partial_d^2 = \frac{1}{2}\partial_z^2$, proving (CD17a). All error bounds include the same field Sobolev factors as (CD17).

Thus an arbitrary pointed carrier may also carry the explicit $(\partial f)X\,DY$ term. It must either be retained or reduced by the exact label/path symmetries and summation by parts of the completed member. It is not discarded by calling the kernel even. Formula (CD17) is the midpoint specialization $z = h/2$.

96.5 5. A sharper coordinate-path estimate by an exact variance identity

For a path consisting of n consecutive positive steps in one coordinate, $h = na$, reflection equals reversal and the area term is zero. In this case the remainder in (CD13) improves to

$$|R_{\mathrm{axis}}| \le C_{M,s} h^{2+2s} \|f\|_{C^{2,2s}} \|X\|_{H^{1+s}} \|Y\|_{H^{1+s}}, \qquad 0 < s < \tfrac{1}{2}. \tag{CD18}$$

This sharper statement is again an actual-lattice estimate.

Let $T = T_{ae_i}$, $G = D_i^U X$, $H = D_i^U Y$, and $A_n = n^{-1}\sum_{k=0}^{n-1} T^k$. Then

$$(T^n - I)X = hA_n G, \qquad (T^n - I)Y = hA_n H. \tag{CD19}$$

For any scalar lattice weight w, bilinearity gives the EXACT variance identity

$$\begin{aligned}(wA_nG, A_nH)_a = \frac{1}{n}\sum_k (wT^kG, T^kH)_a \\ - \frac{1}{2n^2}\sum_{k,l}(w(T^k - T^l)G, (T^k - T^l)H)_a.\end{aligned} \tag{CD20}$$

There is no assumption of independence, positivity, or a Hermitian adjoint in this identity.

Take $w(x) = f(x + h/2)$. Pairing invariance makes the first term

$$\left(\left\{\frac{1}{n}\sum_{k=0}^{n-1} f(\cdot + h/2 - kae_i)\right\} G, H\right)_a.$$

The average sample location is $ae_i/2$, so the weight differs from f by at most

$$C\{a\|\nabla f\|_\infty + h^2\|D^2 f\|_\infty\}. \tag{CD21}$$

In the second term of (CD20), apply (CD10) to both factors. It is bounded by $Ch^{2s}\|f\|_\infty\|G\|_{H^s}\|H\|_{H^s}$. Multiplying by h^2 in (CD19), the weight error in (CD21) is $O(h^2 a + h^4)$, bounded by $O(h^{2+2s})$ because $a \le h$ and $2s < 1$. The central scalar Taylor remainder has the same order when $f \in C^{2,2s}$. Insert these estimates into (CD12). This proves (CD18) uniformly in n and in the microscopic cutoff.

For a general ordered path, the derivative factors in the corresponding telescoping sum point in different directions. The above variance identity does not directly identify its leading cross terms with a single average of one field. Therefore (CD18) is not used for arbitrary paths in the general conclusion.

96.6 6. Exponentially localized kernels and dimension-six scaling

Consider a completed even quadratic carrier family in current units. Its member indexed by Γ has path length $\ell_\Gamma \le rL_\Gamma$, displacement h_Γ, and coefficient matrix k_Γ^{AB}, where A, B index curvature components. There is an algebraic hypothesis in addition to a locality norm: the root sum and the inverse/reflection pairing are completed so that the translation $x \mapsto x + h_\Gamma$ used in (CD12) is a legitimate reindexing with the indicated coefficient. This holds for the full lattice sum with translation-independent k_Γ^{AB}. It also holds for an explicitly completed set of subcell phases and carrier partners with the corresponding exact shifted coefficients.

Summing only over old block-grid roots does not in general have this property when h_Γ is a finer displacement. Likewise an arbitrary root-dependent coefficient $k_\Gamma(x)$ changes under the reindexing. Its difference, or its explicitly justified derivatives, must then be kept in the pointed projection and remainder. Exponential decay in the carrier alone does not bound these coefficient differences. We assume the completed translation pairing here; an equivalent coefficient-difference version needs its own stated norm.

With this algebraic completion, assume the absolute moment bound

$$\sum_{\Gamma,A,B} |k_\Gamma^{AB}|(1 + L_\Gamma)^{2+s} e^{crL_\Gamma} \le K_{2+s}. \tag{CD22}$$

An exponential carrier norm supplies this bound, uniformly for $0 < r \le 1$, after reserving a fixed part of its locality exponent for transports. On a local box, all short paths stay in the protected collar; a long or collar-crossing carrier is retained separately with its original locality reserve. The regular sum considered here contains only carriers whose entire paths lie in a domain with the displayed regular bounds. No claim about a path through a wild hole is made.

Sum (CD17), keeping the actual path areas and all shifted tests. The moments

$$k_0 = \sum_\Gamma k_\Gamma, \qquad k_{ij} = \sum_\Gamma k_\Gamma h_{\Gamma,i} h_{\Gamma,j}, \qquad k_{ij}^{\mathrm{area}} = \sum_\Gamma k_\Gamma \mathcal{A}_{ij}(\gamma_\Gamma) \tag{CD23}$$

are the exact local projection coefficients for this representation. The remainder satisfies

$$|\mathcal{R}_r(f; F)| \le C K_{2+s} r^{2+s} \|f\|_{C^{2,s}} \|F\|_{H^{1+s}}^2. \tag{CD24}$$

Here the curvature cubic terms in (CD17) have also been retained. For rL_Γ beyond a fixed small collar size, use the elementary bilinear bound and the reserved exponential in (CD22), provided that the larger regular domain still contains the path. A carrier leaving that domain is outside this regular estimate and keeps its separate boundary/live norm. The inequality is not proved by applying a small-loop expansion beyond its domain.

The relation to the renormalization scale is worth making explicit. If an old carrier has relative scale r, its own normalized curvature reads as $r^2 F$ in current coordinates. Its quadratic projection starts at r^4, its dimension-six terms are of order r^6, and (CD24) is then of order r^{6+s}. The four-dimensional root count is r^{-4}, leaving r^{2+s}. The dimension-six engineering normalization contributes the remaining r^{-2}, leaving

$$\boxed{r^s.} \tag{CD25}$$

This is a strict geometric age gain. None of these factors is a polynomial loss being hidden in a scale sum.

For example, if a birth member has analytic size b_j on a curvature radius α_j, and its current protected readout has norm $C\alpha_k$, its quadratic contribution after the complete dimension-six projection is

bounded, in the integrated normalized readout, by

$$Cb_j \left(\frac{\alpha_k}{\alpha_j}\right)^2 r^s \|f\|_{C^{2,s}}. \tag{CD26}$$

This formula presupposes the genuine incoming analytic curvature-kernel norm and its admission on the actual prepared graph. It does not manufacture such a norm from a weak bound on the total density.

A local cubic kernel is treated separately. Since $F \in C^\beta$, moving its three curvature arguments and their transports to one root leaves a remainder $O(r^\beta)$ relative to its r^6 leading engineering term. Its complete C-even cubic jet is retained. Terms of curvature degree at least four start at r^8, leaving at least r^2 after the root count and dimension-six normalization, subject to the original analytic radius ratios. These statements require an absolutely summable coefficient series on a common admitted core.

97.1 1. Domain, conventions, and the strengthened CM estimate

Fix an odd blocking factor L. The microscopic spacing is a; the successive spacings are $a_j = L^j a$, with final spacing $a_k = r \le 1$. Work on a periodic lattice, or on a common buffered chart containing every determining path used below. Put

$$U_\mu(x) = \exp\{aA_\mu(x + ae_\mu/2)\}, \qquad M = \|A\|_{C_a^{1,\beta}}, \qquad K = \|A\|_{H_a^{2+s}}, \quad 0 < s < \min(\beta, \tfrac{1}{2}). \quad \text{(CB1)}$$

All Sobolev norms have physical volume normalization. The constants are uniform in a, r/a, and the number of blocking steps. They may depend on L, s, β, the group and a fixed collar. Global K is allowed to scale with the physical volume; a local version uses the corresponding fixed extended carrier.

The buffered version needs no new saddle-extension theorem. On the controlled MC/CL analysis chart, take a fixed smooth cutoff χ equal to one on the UNION OF ALL actual microscopic prefix read sets used by the desired coarse values, and zero near the boundary of a larger regular collar. Extend the potential as χA, then by zero into an auxiliary periodic box. Standard cutoff product estimates preserve H^{2+s} and $C^{1,\beta}$, with constants depending on the fixed collar width in current units. The input link field, every prescribed path, and consequently every intermediate raw mean read by the target carrier are unchanged: this follows inductively from the complete read sets. The auxiliary field outside them is only an analysis extension; it is not asserted to be a critical field or to give the same global constrained saddle. If the available collar shrinks, its cutoff constants must be tracked rather than called uniform.

The field is in the common real or complex logarithmic chart, with $C_L(M+M^2)r$ sufficiently small. For complex fields, the pairing inside observables is the holomorphic invariant bilinear trace pairing; absolute estimates use a matrix norm.

Let

$$A_{j+1} = q^{\mathrm{str}}_{a_{j+1}}(A_j) = Q_{1,j}A_j + N^{\mathrm{str}}_{a_{j+1}}(A_j) \tag{CB2}$$

be the EXACT coarse gauge-straightened actual mean CM17–19. The output connection e^{rA_k} is gauge equivalent to the actual raw iterated mean, by GQ4.

The proof of CM14 extends one Sobolev order. The tame product estimate, uniform on these lattices, is

$$\|uv\|_{H^{1+s}} \le C_s\big(\|u\|_\infty\|v\|_{H^{1+s}} + \|v\|_\infty\|u\|_{H^{1+s}}\big). \tag{CB3}$$

One proof uses physical dyadic difference/Fourier decompositions: the low-frequency factor is bounded in the sup norm and the high-frequency factor carries the $1+s$ weight. The periodic lattice convolution kernels have uniform L^1 bounds, so the constant does not depend on the Nyquist cutoff. Equivalently the integer difference product rule followed by the fractional translation estimate gives the same tame bound. This estimate does not require a second derivative in the sup norm.

For the zero-constant-moment quadratic kernel CM8, finite-L telescoping and sampling give

$$\|S_\zeta A - S_{\zeta'}A\|_{H^{1+s}} \le C_L\ell\|A\|_{H^{2+s}}, \qquad \|S_\zeta A - S_{\zeta'}A\|_\infty \le C_L\ell\|\nabla_a A\|_\infty. \tag{CB4}$$

The degree at least three terms retain their explicit ℓ^2 prefactor. Thus the actual finite kernel satisfies

$$\|N^{\mathrm{str}}_\ell(A)\|_{H^{1+s}_\ell} \le C_{s,L}\ell^2\left[\|A\|_\infty\|A\|_{H^{2+s}_a} + (\|\nabla_a A\|_\infty + \|A\|^2_\infty)\|A\|_{H^{1+s}_a}\right]. \tag{CB5}$$

The identical cancellation and finite-offset argument in C^β gives

$$\|N^{\mathrm{str}}_\ell(A)\|_{C^\beta_\ell} \le C_{\beta,L}\ell^2(M^2+M^3). \tag{CB6}$$

Here the offset difference has C^β norm bounded by $C_L\ell\|\nabla_a A\|_{C^\beta}$.

The flat Q_1 maps contract the physical Bessel norm at every nonnegative order, CM21. They also contract the sup norm of the first difference and its physical Hölder seminorm: the coarse first difference is a positive line average of fine first differences, and a coarse translation is a fine translation of the same physical size. A SINGLE coarse inverse difference in (CB5), respectively (CB6), therefore yields

$$\begin{aligned} \|A_j\|_{C^{1,\beta}_{a_j}} &\le CM, \\ \|A_j\|_{H^{2+s}_{a_j}} &\le e^{C(M+M^2)r}K. \end{aligned} \tag{CB7}$$

Indeed the nonlinear increments are $C\ell_j(M_j+M_j^2)K_j$ and $C\ell_j(M_j^2+M_j^3)$, and $\sum_j \ell_j \le r/(1-L^{-1})$. No per-step norm-equivalence constant is iterated.

Unrolling against the entire flat tail gives, with $E_r = A_k - Q_{1,a\to r}A$,

$$\begin{aligned} \|E_r\|_{H^{1+s}_r} &\le Cr^2(M+M^2)K, \\ \|E_r\|_{H^{2+s}_r} &\le Cr(M+M^2)K, \\ \|E_r\|_\infty &\le Cr^2(M^2+M^3). \end{aligned} \tag{CB8}$$

The fixed small-radius exponential in (CB7) is included in C. These are estimates for the existing actual mean in its explicit analysis gauge.

97.2 2. The positive two-cochain average is exact at the flat level

It is useful to regard each component as an array indexed by its base lattice sites, keeping its geometric midpoint as an associated fixed phase. For a step $\ell = La$, let Q_0 be normalized centered block averaging and let

$$\mathcal{L}_{\mu,L} = L^{-1} \sum_{m=0}^{L-1} \tau_{mae_\mu}. \tag{CB9}$$

In these actual component phases the flat map of CM18 is $Q_{1,\mu} = Q_0 \mathcal{L}_{\mu,L}$. Define

$$Q_{2,\mu\nu} = Q_0 \mathcal{L}_{\mu,L} \mathcal{L}_{\nu,L}. \tag{CB10}$$

This is an analysis operator determined by the ACTUAL flat derivative; it does not replace the nonlinear mean. The elementary identity $(\tau_{\ell e_\mu} - I)/\ell = \mathcal{L}_{\mu,L}(\tau_{ae_\mu} - I)/a$ gives

$$d_\ell Q_1 = Q_2 d_a. \tag{CB11}$$

For any component array Z,

$$D^0_{\rho,\ell} Q_2 Z = Q_2 \mathcal{L}_{\rho,L} D^0_{\rho,a} Z. \tag{CB12}$$

Here D^0 is the flat forward difference. A backward difference has the corresponding shifted positive line filter.

The Q_2 maps, and their whole tails, contract every physical nonnegative Bessel norm. In each alias fibre the centered block factor has squared coefficient sum at most one; each extra line factor has absolute value at most one. Also the coarse Laplacian symbol is bounded by the fine symbol on every alias, as in CM10. Weighted Cauchy–Schwarz proves the contraction. The products over nested steps are the direct block-and-two-line filters, by the finite geometric-sum identity. Thus no loss proportional to the number of intermediate scales occurs in (CB11)–(CB12).

97.3 3. A one-step nonlinear curvature defect, in a differentiated norm

Define the based logarithmic curvature at spacing b by

$$F_{b,\mu\nu}(A)(x) = b^{-2} \log\Big(e^{bA_\mu(x)} e^{bA_\nu(x+be_\mu)} e^{-bA_\mu(x+be_\nu)} e^{-bA_\nu(x)} \Big). \tag{CB13}$$

In this formula only, $A_\mu(x)$ denotes the based component array; its actual link midpoint is $x + be_\mu/2$. The common small branch is understood.

BCH separates the exact linear part:

$$F_b(A) = d_b A + \Gamma_b(A), \qquad \Gamma_{b,\mu\nu}(A) = [A_\mu(x), A_\nu(x)] + \mathcal{R}_{b,\mu\nu}(A). \tag{CB14}$$

The degree-two kernel of $\mathcal{R}_b$ vanishes on constant collections. Subtracting the constant collection with values at the common base point therefore leaves a fine offset difference in every quadratic term. Its displacement is $O(b)$. The terms of degree at least three have an explicit factor b^{n-2}, hence at least one factor b. The ordinary discrete product rule and finite-offset telescoping prove

$$\|\mathcal{R}_b(A)\|_{H^1_b} \le Cb(M + M^2)\|A\|_{H^2_b}. \tag{CB15}$$

For example the derivative of an offset difference is bounded in L^2 by $Cb\|D^2_b A\|_2$, while the undifferentiated offset difference is bounded in the sup norm by CbM. This is exactly where an H^2 norm, but

not $\|D^2A\|_\infty$, is used. The entire Γ_b has uniformly bounded analytic tame product norms on the same branch.

For one fixed-L step, compare $\Gamma_\ell(Q_1A)$ and $Q_2\Gamma_a(A)$. Choose one fine offset for each component in the determining patch. Each of Q_1A_μ minus that sample, and each sample difference occurring in $Q_2[A_\mu, A_\nu]$, has sup norm at most $C_L\ell M$ and H^1 norm at most $C_L\ell\|A\|_{H^2_a}$. The fixed-L sampling estimate CM11 controls those sampled norms. Subtract and add the commutator of the same two samples. The product rule, followed by (CB15), gives

$$\|\Gamma_\ell(Q_1A) - Q_2\Gamma_a(A)\|_{H^1_\ell} \le C_L\ell(M + M^2)\|A\|_{H^{2+s}_a}. \tag{CB16}$$

All offsets and midpoint phases remain those of the actual one-step map. The argument is only at fixed L; no sparse sampling estimate with a growing r/a has been inserted.

Now put $A' = q^{\mathrm{str}}_\ell(A) = Q_1A + N^{\mathrm{str}}_\ell(A)$. The linear curvature difference is $d_\ell N^{\mathrm{str}}_\ell$ by (CB11), and its H^1 norm is at most $C\ell^{-1}\|N^{\mathrm{str}}_\ell\|_{H^1} \le C\ell(M + M^2)\|A\|_{H^{2+s}}$. The analytic product bound for $\Gamma_\ell(Q_1A + N) - \Gamma_\ell(Q_1A)$ is smaller, of order ℓ^2 times the same type of norm. Thus

$$\boxed{\|F_\ell(q^{\mathrm{str}}_\ell A) - Q_2F_a(A)\|_{H^1_\ell} \le C_L\ell(M + M^2)\|A\|_{H^{2+s}_a}.} \tag{CB17}$$

Apply this identity to each actual intermediate A_j. Unroll the curvature recurrence against the ENTIRE Q_2 tail, using (CB7) and the H^1 contraction. It follows that

$$\boxed{\|F_r(A_k) - Q_{2,a\to r}F_a(A)\|_{H^1_r} \le Cr(M + M^2)K.} \tag{CB18}$$

The norm includes one derivative. Differentiating this error therefore costs NO r^{-1} factor.

The direct local analytic formula (CB13), (CB7), and the tame product bound also give

$$\|F_{a_j}(A_j)\|_{H^{1+s}_{a_j}} \le C(1 + M)K, \qquad \|F_{a_j}(A_j)\|_{C^\beta_{a_j}} \le C(M + M^2). \tag{CB19}$$

These are the regular coarse curvature bounds needed to run CD on the actual own coarse lattice.

97.4 4. Lift an already differentiated coarse field, never differentiate a hold

Let $\mathcal{P}_{r\to a}$ hold a coarse component constant on its corresponding centered fine cells, with the correct component phase. It is an L^2 isometry in physical volume normalization. The composition of this hold with Q_2, or with Q_2 and one more positive line filter, has a positive kernel $K_r(x, y)$ with

$$\sum_y K_r(x, y) = 1, \qquad K_r(x, y) = 0 \text{ if } |x - y| > Cr, \qquad K_r(x, y) \le C(a/r)^4. \tag{CB20}$$

Its column sums are bounded as well (indeed the full periodic filters preserve physical mass). For $0 < \sigma \le 1$, Jensen and the translation-difference characterization of the discrete Sobolev norm imply

$$\|\mathcal{P}_{r\to a}Q_2Z - Z\|_{2,a} \le Cr^\sigma\|Z\|_{H^\sigma_a}. \tag{CB21}$$

To see the uniformity directly, square the left side and bound it by

$$C(a/r)^4 \sum_{\substack{h\in a\mathbb{Z}^4\\ |h|\le Cr}} \|\tau_hZ - Z\|^2_{2,a}. \tag{CB22}$$

Each summand is at most $C|h|^{2\sigma}\|Z\|^2_{H^\sigma_a}$, and the normalized number of shifts is bounded. The same proof applies to the extra line filter in (CB12), including backward and shifted versions.

Combining (CB12), (CB18), and (CB21), with $\sigma = s$, gives

$$\|\mathcal{P}_{r\to a}D^0_{\rho,r}F_r - D^0_{\rho,a}F_a\|_{2,a} \le C\left[r^s\|F_a\|_{H^{1+s}_a} + r(M+M^2)K\right]. \tag{CB23}$$

This is NOT an H^1 bound for the held field $\mathcal{P}_{r\to a}F_r$. Differentiating a sharp hold on the microscopic lattice would introduce artificial interface jumps and would not prove (CB23).

97.5 5. Flux and covariant directional-current comparison

The actual Wilson flux is

$$X_b = b^{-2}\pi\frac{P_b - P_b^{-1}}{2} = \Phi_b(F_b), \qquad \Phi_b(Z) = b^{-2}\pi\sinh(b^2Z). \tag{CB24}$$

Here π is the holomorphic traceless projection, agreeing with the compact-group convention on the real slice. On the controlled branch, $\Phi_b(Z) = Z + b^4O(Z^3)$. Its tame product estimates show

$$\|X_b - F_b\|_{H^1_b} \le Cb^4\|F_b\|^2_\infty\|F_b\|_{H^1_b}, \qquad \|X_b\|_{H^{1+s}_b} \le C\|F_b\|_{H^{1+s}_b}. \tag{CB25}$$

Consequently (CB18) and (CB23) hold for X, with fixed polynomial modifications of the M-dependent constants.

The exact covariant forward difference is

$$D^{A_b}_{\rho,b}X_b = D^0_{\rho,b}X_b + \mathcal{A}_{\rho,b}(A_b)\tau_{be_\rho}X_b, \qquad \mathcal{A}_{\rho,b}(A_b) = b^{-1}(\mathrm{Ad}_{e^{bA_{b,\rho}}} - I). \tag{CB26}$$

It obeys $\|\mathcal{A}_{\rho,b} - \mathrm{ad}\,A_{b,\rho}\|_\infty \le CbM^2$. Positivity and finite range of Q_1, together with (CB8), give the actual potential comparison

$$\|\mathcal{P}_{r\to a}A_k - A\|_\infty \le CrM + Cr^2(M^2+M^3). \tag{CB27}$$

Thus the lifted coarse coefficient in (CB26) differs from the fine one by $O(r(M+M^2))$. The shifted flux factors differ in L^2 by $O(r)$ times their H^1 bounds, using (CB18), (CB21) with $\sigma = 1$, and fine translations. The backward difference has the same argument with its exact backward coefficient.

Let $K_* = C(1+M)^C K$, and $M_F = C(M+M^2)$. We have proved

$$\begin{aligned}
\|\mathcal{P}_{r\to a}X_r - X_a\|_{2,a} &\le CrK_*,\\
\|\mathcal{P}_{r\to a}D^{A_k}_{\rho,r}X_r - D^A_{\rho,a}X_a\|_{2,a} &\le Cr^sK_*,\\
\|\mathcal{P}_{r\to a}(D^{A_k}_{\rho,r})^*X_r - (D^A_{\rho,a})^*X_a\|_{2,a} &\le Cr^sK_*.
\end{aligned} \tag{CB28}$$

The stars here have the exact invariant-pairing convention $(T^{-1}-I)/b$, not a derivative of a complex conjugate background. The connection terms are treated explicitly; they are not replaced by flat derivatives.

97.6 6. Comparison of the actual tested H and $C^\sharp$

At spacing b, set

$$J^{(b)}_{\nu;\mu} = (D^{A_b}_{\mu,b})^*X_{b,\mu\nu}, \qquad Q^{(b)}_{\nu;\mu} = J^{(b)}_{\nu;\mu} - \tfrac{1}{3}\sum_{\lambda\neq\nu}J^{(b)}_{\nu;\lambda}, \qquad H_b(f) = b^4\sum_{x,\nu,\mu\neq\nu}f(x)\langle Q^{(b)}_{\nu;\mu}, Q^{(b)}_{\nu;\mu}\rangle. \tag{CB29}$$

This is the actual directional-current complement AI33–35. Its fixed hypercubic symmetrization satisfies the same bounds. For a smooth physical test f, holding its coarse samples changes its fine samples by at most $Cr\|\nabla f\|_\infty$. The finite projection $J \mapsto Q$, physical L^2 isometry of the hold, and (CB28) imply

$$\boxed{|H_r(f; e^{rA_k}) - H_a(f; U)| \le Cr^s \|f\|_{C^1} K_*^2.} \tag{CB30}$$

Every source shift in a fixed reflection convention is kept until this test comparison; it costs at most the displayed $r\|\nabla f\|_\infty$ term.

For C_b^{split}, use the exact transported cubic AI24. Each of its paths has a fixed number of lattice steps. Moving such a transported flux to its base point costs at most $Cb\|D_b X_b\|_2$, with bounded short transports. Alternatively compare the exact shifted factors using (CB28). With $\|X_b\|_\infty \le CM_F$, telescoping the three factors and comparing the test samples gives

$$|C_r^{\mathrm{split}}(f; e^{rA_k}) - C_a^{\mathrm{split}}(f; U)| \le Cr\|f\|_{C^1} M_F K_*^2. \tag{CB31}$$

The difference between $C^\sharp$ and C^{split} must also be retained. The exact cube identity AI25–27 gives on this regular branch

$$\|B_b\|_2 \le CbM_F\|F_b\|_2, \qquad |R_{4,b}(f)| \le Cb^2\|f\|_\infty M_F^2 \|F_b\|_2^2. \tag{CB32}$$

The first estimate follows by bounding one of the two plaquette defects in every cube product by $Cb^2 M_F$; the other is estimated in L^2. The finitely many transports are bounded. The same polynomial identities extend holomorphically from the real group to the common complex chart, with fixed absolute constants.

In the tested form of AI32a–b, move the last scalar difference onto f by EXACT lattice summation by parts. It follows that

$$|C_b^\sharp(f) - C_b^{\mathrm{split}}(f)| \le C\left[b^2\|f\|_\infty M_F^2 + b\|\nabla f\|_\infty M_F\right] K_*^2. \tag{CB33}$$

No pointwise high derivative of B_b has been assumed. Combining (CB31)–(CB33), with $a \le r$, proves

$$\boxed{|C_r^\sharp(f; e^{rA_k}) - C_a^\sharp(f; U)| \le Cr\|f\|_{C^1}(M_F + M_F^2)K_*^2 \le Cr^s\|f\|_{C^1}(M_F + M_F^2)K_*^2.} \tag{CB34}$$

This controls the particular fixed compact completion $C^\sharp$, rather than silently replacing it by a different cubic regulator. The explicit factor M_F is retained: on a canonical family with $M, K = O(g)$, this comparison is of cubic order $O(rg^3)$. Replacing that factor by a constant would lose the coupling grade needed for a normalized cubic source.

For the quadratic descendant $\mathcal{Q}_b(\Delta_b f) = b^4 \sum (\Delta_b f)|X_b|^2$, the central scalar difference formula gives $\|\Delta_b f - \Delta f\|_\infty \le Cb^s\|f\|_{C^{2,s}}$. The flux comparison and test quadrature therefore give

$$|\mathcal{Q}_r(\Delta_r f; e^{rA_k}) - \mathcal{Q}_a(\Delta_a f; U)| \le Cr^s\|f\|_{C^{2,s}} K_*^2. \tag{CB35}$$

The same statement holds for the individual second test derivatives retained by the full tensor projection. This does not remove EOM/contact pushforwards: those remain the exact separate module of AI.

For clarity, the bridge holds before the scalar quotient is taken. Equation (CB28) compares every component of the full derivative-curvature quadratic tensor:

$$\left| r^4 \sum f\langle D_{\rho,r} X_{r,A}, D_{\sigma,r} X_{r,B}\rangle - a^4 \sum f\langle D_{\rho,a} X_{a,A}, D_{\sigma,a} X_{a,B}\rangle \right| \le Cr^s\|f\|_{C^1} K_*^2. \tag{CB35a}$$

Here A, B label arbitrary curvature components and each displayed derivative is the actual covariant difference, with either retained forward/backward convention. Likewise the extra pointed moment in CD17a obeys

$$\left| r^4 \sum (\partial_\lambda f) \langle X_{r,A}, D_{\rho,r} X_{r,B} \rangle - a^4 \sum (\partial_\lambda f) \langle X_{a,A}, D_{\rho,a} X_{a,B} \rangle \right| \le C r^s \|f\|_{C^2} K_*^2. \tag{CB35b}$$

Proof is one flux difference, one derivative difference, and the test quadrature estimate. The fixed finite shifts of the actual source-anchor convention are retained in the same way. All cubic path-area contractions with fixed bounded invariant coefficients have the estimate (CB31), including its factor M_F. Thus none of these pointed tensor terms is removed merely by passing to the two-dimensional scalar quotient.

98.1 1. Identify the source slot before changing its extension

Let $\Phi_j(U, J, B)$ denote a **completed** original minimum readout in a single RG step. Its defining physical identity is

$$\Phi_j(U_{j+1}(V), J_{\mathrm{phys}}(U_{j+1}(V)), B) \sim U_j(V_j(B, V)). \tag{PA1}$$

The right side is the actual minimum at level j with its actual determining data and boundary class. The symbol $\sim$ permits only the original gauge change, which disappears in a gauge-invariant insertion.

This is the type of completed slot used in I (3.3), (3.18)–(3.19), and (3.33)–(3.36). In the near-window construction an outer interpolated potential is first converted into determining data, and an **inner** U_j is reconstructed before the old function is read. The arbitrary outer field $e^{gB}U$, or $tH(B)$, is not itself declared to satisfy (PA1). An actual Haar source in an uncompleted live operator is also not a slot of type (PA1).

Write $\mathcal{R}_{j,\mathbf{1}}$ for the new MC reconstruction at all walls one, on the same global hierarchy, action, and allowed boundary class. It has the exact identity

$$\mathcal{R}_{j,\mathbf{1}}(W) = W \quad \text{on the admitted level-}j\text{ critical graph.} \tag{PA2}$$

For an off-graph paired argument define

$$\widehat{U} = \mathcal{R}_{j+1,\mathbf{1}}(U), \qquad \widehat{J} = J_{\mathrm{phys}}(\widehat{U}). \tag{PA3}$$

In particular the current in this auxiliary source construction is the physical current of the repaired background. It is not left as an arbitrary independent J when invoking a theorem about a physically regular nonlinear response.

For a gauge-invariant primitive $\mathcal{O}_j$, a source extension is

$$\mathcal{O}_j^\sharp(U,J,B)=\mathcal{O}_j\Big(\mathcal{R}_{j,\mathbf{1}}\big(\Phi_j(\widehat{U},\widehat{J},B)\big)\Big). \tag{PA4}$$

Equation (PA4), including its inner repair and physical current, is the chosen paired-core definition in this note and the version used when MT cites PA4. A post-repair-only definition can also be used on the explicitly verified original response class described at the end of Section 5. These are two allowable extensions; they are not silently identified away from the physical graph. The outer reconstruction is redundant on a genuine critical Φ_j; it is useful when the source maps themselves are localized or when only a low-core range bound has been imported. There are a fixed number of reconstructions in (PA4), independent of the age of an inherited activity.

On every physical input in (PA1), (PA2) first removes the inner repair of U, and then the outer repair of the completed level-j minimum. Therefore, POINTWISE in every admitted fluctuation value B,

$$\mathcal{O}_j^\sharp(U_{j+1}(V),J_{j+1},B)=\mathcal{O}_j(U_j(V_j(B,V))). \tag{PA5}$$

Consequently this substitution in a source slot preserves its physical integral, without changing the Gaussian covariance, coarea/Jacobian factor, real cutoff, conditional numerator, or normalizing denominator. This is a pointwise source identity; it is not a change of integration coordinates.

98.2 2. Why the original low core is enough for the input

The original first-three-clause paired core controls a real regular reference, a small complex first-order deformation, and small curvature/current. On each fixed complete cell carrier its reference has a low analysis chart; the deformation is C^1, hence $W^{1,p}$. Its explicit curvature bound supplies the separate curvature condition in MC2. Thus, after a fixed parameter choice,

$$\|(U,J)\|_{\text{original low core}}\le c_0 \quad\Longrightarrow\quad U\in\text{the admitted MC weak/curvature tube.} \tag{PA6}$$

This implication uses the fixed geometry constants and the smaller original field radii. It is not a uniform assertion for a supremum link tube without first-order control.

The output $\widehat{U}$ has the protected low and fractional curvature/current bounds. The original full nonlinear relative response has a separate primary input. CMP 102 (1985), Proposition 9, printed p. 309, asserts analyticity of its full response in the coarse perturbation B and external regular configuration. Equations (173)–(180) give its expansion with linear first term; the weighted regularity conditions (19) include the covariant Laplacian, and (190) bounds its first functional derivative with spatial decay.

Accordingly, on a fixed smaller physical B ball and a fixed smaller external regular core, this response has a uniform first-order field/curvature range bound. In symbols,

$$\|\Phi_j(\widehat{U},\widehat{J},B)\|_{\text{low tube}}\le C_{\text{geom}}\{\|\widehat{U}\|_{\text{low tube}}+\|B_{\text{physical}}\|_\infty\}. \tag{PA7}$$

Here the coarse perturbation uses its actual physical normalization. The absolute field radii and physical fluctuation cutoff must be chosen so that the right side lies in the second MC tube. This is one fixed compatibility condition, not a radius loss at every later age.

I (3.14)–(3.17) keeps the original Laplacian-response bounds under its later localization in the remote/large-carrier part. Its near-window part, (3.31)–(3.36), first allows a merely first-order regular outer potential and then completes the inner minimum. These are different constructions. Neither statement licenses replacing the full physical curvature derivative of an arbitrary independent-current background by its independent variable J.

On the common admitted domain, MC, the original response, and the physical current are holomorphic functions of their actual field coordinates. Thus (PA4) is a holomorphic source extension for each fixed cutoff. Uniform bounds for derivative insertions come from its protected fractional output, rather than from a pointwise high-jet bound on the arbitrary original input.

The same discussion applies on a mixed regular hierarchy only where its actual layer bounds, boundary conditions, and complete read sets are included. An unresolved rough source factor remains in its original joint cold operator; it is not fed into (PA4).

98.3 3. The protected collar must be part of the source carrier

For exact physical equality, the reconstruction in (PA4) is first defined globally on the SAME determining problem. This can itself be an original finite-window problem with its actual prescribed fine boundary data; it need not be the entire physical torus. In that case the MC action graph, pinned collar, and fixed exterior coordinates must be those of that very window problem. Its localization then uses the common typed carrier expansion. Keep unweakened a fixed enlargement of the primitive support containing:

- its actual plaquettes, covariant differences, and test shifts;
- every differential stencil needed by the protected equations;
- every actual averaging/prefix read set and incidence stencil entering that collar.

Outside this fixed source collar the new wall variables can be expanded. All resulting source carriers include the collar and the actual connected paths of that expansion. Their extra base volume is a fixed geometry cost. The low source input range and protected output bounds hold on a common wall polydisc by MC. The original formation wall variables may be kept as separate variables and their carriers joined with these source carriers. At all of them equal to one, (PA5) is exact.

An independent periodic extension of the physical input, followed by a local saddle solve, is NOT interchangeable with this construction. It has the same local input on an inner box but need not be the same critical field there. Exponential closeness would not prove the source identity (PA5).

The finite local factors of the common parametrix may use their own CE extensions and periodic inverses, as already proved in MB/MC. Those extensions define the individual factors in an EXACT expansion of the global map; they are not a replacement of that global endpoint by one local saddle.

For a localized source coefficient, every microscopic read set of those factors is included in its completed carrier. The coefficient is then defined from those finite local operations on that carrier. This is the new source construction's naturality; it does not identify the old unmarked source-zero vacuum factors.

98.5 5. A genuine canonical radial disc inside the original domain

One need not assume that the whole original paired space is star-shaped. There is a smaller, explicitly parameterized subclass where the needed complex t disc is justified. On a fixed completed carrier choose a common identity chart and write

$$U_t = e^{atZ}, \qquad J_t = t\,j, \qquad B_{\mathrm{physical},t} = tB, \qquad q_\infty = 1 + \|Z\|_{C^1} + \|j\|_\infty + \|B\|_\infty. \tag{PA10}$$

Use the actual coarse fluctuation normalization; analytic bounded linear covariances and the original quadratic coordinate correction can be included in $B_{\mathrm{physical},t}$ with a fixed increase of the constants. This is a FLAT identity-chart family for the entire determining problem used by Ψ_t: every prescribed collar, exterior, reference, older determining, and pinned boundary coordinate that the source slot reads must equal its identity value at $t = 0$. If such coordinates are external field variables, they are included among the coordinates scaled by t, and their low norms enter q_∞. Fixed geometry, tree choices, and source-free cutoff parameters need not be scaled. A frozen nonflat exterior is not covered by the assertion $\Psi_0 = 1$; its contribution belongs to the separate full physical-core estimate (PA9), or remains inside the actual joint cold operator. Scaling only the displayed inner U, J, B while holding a nonflat determining collar fixed would not justify the canonical expansion below.

For $|t|q_\infty < c$, choose the real reference to be the identity. The complex deformation is tZ, whose potential and first differences are bounded by c. Exact plaquette telescoping gives

$$\|F_a(U_t)\|_\infty \le C|t|\|DZ\|_\infty + C|t|^2\|Z\|_\infty^2 \le Cc, \qquad \|J_t\|_\infty \le c. \tag{PA11}$$

Take c below the actual smaller-core inclusion constant in I following (1.16). The first three clauses then imply the full original paired-domain clause (iv), uniformly along this entire complex disc. Thus (PA10) is not an assumed radial interpolation through arbitrary points of a non-star-shaped domain.

The original response and near-window reconstruction are analytic and uniformly low-regular on that smaller domain. For the complete flat determining family just specified, uniqueness of the zero-data reconstruction makes them the identity at $t = 0$ in their actual analytic Landau/reference charts. Their uniform low bounds, followed by one-variable Banach-valued Schwarz/Cauchy on the disc c/q_∞, therefore give for the complete source slot

$$\|\Psi_t - 1\|_{\mathrm{MC,input}} \le C|t|q_\infty \quad \text{on } |t|q_\infty < c/2. \tag{PA12}$$

The norm in (PA12) includes the first-order relative coordinates and the separate curvature bound, not an inferred high derivative of the original background. Changes of its controlled local analysis frame use the intrinsic covariant norms of MC. A coarse canonical background may instead be parameterized by its actual minimum $U_{j+1}(e^{tz})$; Proposition 9 supplies the same disc and all nested conditions directly.

This proves the input step of MT22 for this explicit canonical class. At the physical point $t = g$, it requires $gq_\infty < c/2$, which follows from the actual pointwise physical cutoff and fixed canonical external ball after fixing their margin. The supremum/row-sum response bounds (19), (173), and (190) are uniform in the carrier size. A local energy $(\sum_{X^+} |B|^2)^{1/2}$ must NOT replace q_∞ when asserting eligibility for this disc: it can grow as $\sqrt{|X|}$ while the pointwise cutoff is unchanged.

With a protected MC output, its zero value at $t = 0$ and its uniform bound on the same disc give curvature $O(|t|q_\infty)$ in the required local H^{1+s} and supremum norms. The actual H and $C^\sharp$ small-field valuations then give $O(|t|^2 q_\infty^2)$ and $O(|t|^3 q_\infty^3)$, respectively. Dividing by t^d and applying Cauchy proves the canonical nonlinear difference $O(gq_\infty^{d+1})$ on a fixed smaller disc. Only AFTER this estimate, on a

finite completed carrier, use

$$q_\infty^{d+1} \le C_d \left(1 + \|Z\|_{C^1} + \|j\|_\infty + \sqrt{\sum_{b \in X^+} |B_b|^2} \right)^{d+1} . \tag{PA13}$$

The joint Gaussian moment now costs a fixed polynomial in the carrier volume, payable from spare locality. This does not shrink the analytic disc. On an infinite carrier one instead uses an actually proved localized kernel/energy decomposition before taking moments, not the global supremum of an infinite Gaussian. This is a valid canonical-center estimate; it does not assert uniform $O(g)$ control on the whole fixed physical core.

The post-MC version MT21 is therefore sufficient whenever the original completed Ψ_t has the low range used in (PA12) on this exact admitted class. The pre-repair in (PA3), with the actual repaired current, is an optional alternative when one wishes to invoke the physical nonlinear-response theorem directly. Neither option changes the physical equality (PA5). The pre-repair does not improve the unavoidable full-core g^{-d} growth in (PA9).

99.1 1. Which original function is being bounded

Write $g = g_k > 0$, and let E_k denote the complete inherited effective interaction in the convention of I (1.6). In particular, the scalar Wilson counterterm $-\beta_j A$ stays paired with its birth-j contribution. The inherited term in the exponent of I (2.12)–(2.13) is

$$\mathcal{D}_k(U, J, B) = E_k\big(\Phi_k(U, J, B)\big) - E_k\big(\Phi_k(U, J, 0)\big) \tag{IV1}$$

on the physical graph, followed by the particular paired-core extension constructed in I Sections 3–5. This notation records the complete original readouts; it is not a definition by a different reconstructed critical map.

The identities needed to identify the counterterm convention are I (0.23), (0.28), and (4.38)–(4.45), followed by (5.37)–(5.43). The quadratic polarization coefficient and its nonlinear gauge completion cancel against the corresponding terms from the negative scalar Wilson counterterm. These cancellations occur before II localizes the resulting expressions. II Section 1, first paragraph, explicitly starts from these already-cancelled terms. Thus neither an unpaired old $E^{(j)}(X)$ nor a separately estimated $\beta_j A$ is the function to which the strict age bound is applied.

II Lemma 1, (1.33)–(1.36), then supplies the exact representation

$$\mathcal{D}_k(U, J, B) = \sum_Y V_k'(Y; U, J, B), \qquad |V_k'(Y; U, J, B)| \le C_* E_0 \epsilon_1 e^{-\kappa_* d_k(Y)} \tag{IV2}$$

on the common domain

$$(U, J) \in \mathcal{U}_{k+1}^{\mathbb{C}}\big(Y, (1+\beta)\alpha_0, (1+\beta)\alpha_1, \alpha_0\big), \qquad \|B\|_{\infty,Y} < \epsilon_1/g. \tag{IV3}$$

Here C_* includes the fixed M, κ_1, and field-radius constants displayed in (1.36). They are fixed once and do not depend on k, the ultraviolet depth, or the current carrier. The positive κ_* is the locality exponent remaining after the original sums. In particular, the old-root and old-age sums have ALREADY been performed in (IV2).

99.2 2. Vanishing is preserved coefficientwise by the actual walls

The improvement of (IV2) requires the identity

$$V_k'(Y; U, J, 0) = 0 \quad \text{for every } Y. \tag{IV4}$$

Vanishing of the sum alone would not suffice. The actual original construction proves (IV4), as follows.

For every complex wall vector s in the common domain, the nonlinear response used by II is

$$\mathcal{H}_{k,s}(B') = H_{0,s}B' + A_{0,s}(B') - H_s D_s(H_{0,s}B' + A_{0,s}(B')). \tag{IV5}$$

This is II (1.17), with its arguments shortened. The two quantitative fixed-point estimates (1.14) and (1.16) give

$$D_s(A') = O(\|A'\|^2), \qquad A_{0,s}(B') = O(\|B'\|^2). \tag{IV6}$$

Consequently $\mathcal{H}_{k,s}(0) = 0$, for each s, independently of whether the external pair (U, J) lies on a physical critical graph. This is a property of the actual already-extended response formula. The local fluctuation-coordinate map II (1.19) is

$$B' = gCB - hD(gCB), \qquad B'(0) = 0. \tag{IV7}$$

For the near-window terms, II (1.1), (1.8)–(1.10), and (1.23) first substitute

$$A = (t\zeta + t_D\zeta_D)\mathcal{H}_{k,s}(B') \tag{IV8}$$

into the appropriate analytic function, then differentiate with respect to t_D at zero, and take the wall derivatives. At $B = 0$ the substituted argument in (IV8) is identically zero for all t, t_D, s. Hence its t_D derivative is identically zero there. Wall differentiation, evaluation at zero walls, and the FTC integrals preserve that zero identity. Equivalently, if $F(s, t_D, 0)$ is independent of t_D, then $\partial_s^\nu \partial_{t_D} F(s, t_D, 0) = 0$.

The remote/large-carrier terms are the I (3.7), (3.15) expressions treated separately in II after (1.29), together with the last integral in I (3.21). They carry the distinguished variation δH_j induced by $\mathcal{H}_{k,s}(B')$, or the corresponding interpolation derivative. It is zero at $B = 0$. The same observation applies to their additional localized response factors: derivatives of H_j may be nonzero at zero, but the distinguished variation multiplying them is zero.

The Ward rewriting does not introduce an unpaired constant. I (4.6) starts with a distinguished δB factor. The algebraic identities, summation by parts, and momentum remainder decomposition retain that variation or a derivative of it. The Wilson-counterterm differences have the same property. Thus their cancellation and regrouping preserve the zero identity before the subsequent wall expansion.

Finally, the original absolutely convergent sums over the old origin, carrier, and localization choices can be taken at $B = 0$. They therefore preserve this identity separately in each final carrier Y. This proves (IV4).

This argument concerns the complete local functions produced by the actual construction. It does not assert that an arbitrary decomposition of a function vanishing at $B = 0$ has vanishing individual pieces.

99.3 3. Radial Schwarz gives the coupling factor without a cutoff loss

Fix an admitted (U, J), a carrier Y, and $b = \|B\|_{\infty,Y} > 0$. The scalar function

$$f(z) = V_k'(Y; U, J, zB) \tag{IV9}$$

is holomorphic on the disc $|z| < R$, where $R = \epsilon_1/(gb)$. It satisfies $f(0) = 0$ by (IV4) and $|f(z)| \le C_* E_0 \epsilon_1 e^{-\kappa_* d_k(Y)}$ by (IV2). The ordinary radial Schwarz lemma, or Cauchy's formula for $f(z)/z$, therefore gives, at every admitted B,

$$|V_k'(Y; U, J, B)| \le C_* E_0\, g\, \|B\|_{\infty,Y} e^{-\kappa_* d_k(Y)}. \tag{IV10}$$

For $B = 0$ the inequality is exact. No star-shapedness of the paired (U, J) domain is used: only the literal product ball in B from II (1.34) is scaled. In particular, the external background and its determining collar may be nonflat.

II, printed p. 8, immediately after (1.29), explicitly remarks that $g_k|B|$ can be retained in place of ϵ_1. This is independent primary confirmation of the same gain. The argument above identifies why it survives in each actual localized inherited term and after the age summation.

Only after obtaining (IV10) introduce a finite-carrier energy,

$$\mathcal{E}_Y(B) = \sum_{b\in Y} |B_b|^2, \qquad |V_k'(Y;U,J,B)| \le g\, v_Y (1+\mathcal{E}_Y(B))^{1/2}, \quad v_Y = C_* E_0 e^{-\kappa_* d_k(Y)}. \tag{IV11}$$

The B-disc was controlled by the supremum, not by $\sqrt{\mathcal{E}_Y}$. Thus a large carrier does not reduce the physical cutoff margin. The one-joint-kernel Gaussian moment costs a fixed polynomial in $|Y|$; a spare part of the exponential locality pays it. If a later linear Gaussian normalization is localized, it preserves zero at $B = 0$. Its proved uniform row bounds and enlarged completed carriers give the same statement with fixed modified constants.

Choosing any final locality exponent strictly below κ_*, with the usual polymer-count margin, gives

$$\|v\|_\mu \le C E_0. \tag{IV12}$$

This is the inherited part of MT12, with polynomial degree one. It is stronger than the proposed interpolation of an $r^{4+\alpha}$ estimate with an $r^4 g^2$ estimate.

100.1 1. Conventions and the two source degrees

Use AI's fixed globally smooth, symmetrized compact representatives, including their pointed test conventions. In current length units write

$$\mathcal{O}_{H,g}(f) = g^{-2} H_a(f), \qquad \mathcal{O}_{C,g}(f) = g^{-3} C^{\sharp}_a(f). \tag{MT1}$$

These powers distinguish canonical field degree from engineering dimension. Both operators have engineering dimension six. H has ordinary field valuation two and C-sharp has valuation three. Thus their integrated engineering transfer is the same, although their canonical coupling normalizations differ. A change of the running g between adjacent scales contributes the diagonal factors (g_{j+1}/g_j)^2 and (g_{j+1}/g_j)^3. On the stated running trajectory these are I+O(g_j^2).

The lower state contains the vacuum, the full quadratic curvature family, its pointed descendants, and the exact EOM/contact module AI9–13. The two-dimensional quotient only appears after these are retained. In particular C-sharp is C even; no C-odd cubic projection is used.

100.2 2. A genuine obstruction on the unmodified paired core

This example concerns a core which bounds a background potential and its first covariant difference, its curvature, and an INDEPENDENT current argument. It does not concern a core which already bounds the actual derivative of curvature or requires the current argument to equal the physical Wilson current.

Take SU(2), choose an anti-Hermitian Cartan generator T with -Tr(T^2)=1, and take a periodic torus of physical side length t_0 for which t_0/a is even. This is compatible with odd blocking L by taking an even number of retained cells. Put

$$A_2(x) = \alpha a (-1)^{x_1/a} T, \qquad A_\mu = 0 \quad (\mu \neq 2), \qquad U_\mu(x) = \exp(a A_\mu(x)). \tag{MT2}$$

Then

$$\|A\|_\infty = \alpha a, \quad \|DA\|_\infty = 2\alpha, \quad F_{12}(x) = -2\alpha(-1)^{x_1/a}T, \quad \|F\|_\infty = 2\alpha. \tag{MT3}$$

All other plaquette logarithms vanish. All link and plaquette charts are on their common small branch for α small. The independent current argument can be chosen identically zero. Thus every displayed first-order background bound remains uniformly small.

The actual traceless Wilson flux is instead

$$X_{12}(x) = -b_a(-1)^{x_1/a}T, \qquad b_a = \frac{\sqrt{2}}{a^2}\sin(\sqrt{2}\,\alpha a^2). \tag{MT4}$$

Only the plane-1 contribution to the physical current in direction 2 is nonzero; its magnitude is 2b_a/a. Subtracting one third of the total current in each of its three plane slots gives

$$H_a(1) = \frac{8}{3}\,\frac{b_a^2}{a^2}\,t_0^4 \sim \frac{32}{3}\,\frac{\alpha^2}{a^2}\,t_0^4. \tag{MT5}$$

The equality uses the physical counting measure a^4 and the identity $(2/3)^{2+(-1/3)}2+(-1/3)\hat{}2=2/3$. Symmetrizing the pointed scalar representative does not change this constant-test value.

Setting α=g ϵ proves that g$\hat{}${-2}H_a cannot have a cutoff-independent bound by any fixed polynomial in 1+Gaussian-energy at zero fluctuation on that unmodified core. The problem occurs before any near-mark integration or mixing. This example is not a physical constrained critical orbit: its actual current grows like alpha/a. It is not a counterexample to MC or to a quantum construction.

100.3 3. Linear current-unit Gaussian fluctuations have a better estimate

The background obstruction is distinct from the linear reconstructed fluctuation. Suppose the actual source-free current-unit response kernel satisfies, on a complete unit carrier Q and its original regular background,

$$(|\mathcal{H}| + |\nabla_U\mathcal{H}| + |\Delta_U\mathcal{H}|)(Q,y) \le Ce^{-c\,\mathrm{dist}(Q,y)}. \tag{MT6}$$

This is the A/first-difference/Laplacian kernel bound described in I (3.14), with the localized replacements in (3.16)–(3.17). For current-unit canonical coordinates B_y, Cauchy–Schwarz in the exponential kernel gives the corresponding local L2 bounds for $v = \mathcal{H}B$, its first derivative, and its covariant Laplacian. In a controlled original C^1 chart the ordinary/covariant Laplacian difference has bounded zero- and first-order coefficients. The cutoff-uniform local lattice elliptic estimate therefore yields

$$\|v\|^2_{H^2(Q)} \le C\sum_y e^{-c'\mathrm{dist}(Q,y)}|B_y|^2. \tag{MT7}$$

This proves a single quadratic energy envelope for the first derivative of the LINEAR fluctuation curvature. No second derivative of curvature is asserted. The actual canonical coordinates in this statement live on the current retained lattice; they are not an unrenormalized collection of all fine ultraviolet coordinates.

The nonlinear relative response has additional primary support: CMP102, Proposition 9, printed p.309, states analyticity of the full response and its bounds (19)–(21), including the covariant Laplacian, with

its derivative estimate (190). Equations (174)–(177) display its linear part and degree-two remainder. Banach-valued Cauchy on a smaller admitted response ball consequently controls that remainder in the same Laplacian norm. This fact does not by itself bound D F_U for an arbitrary outer background U. Nor does it identify every independent-current extension with the physical minimum response.

100.4 4. Polynomial marks in one connected formation

Here is an abstract lemma whose inputs are stated before the marked output is formed. Use exactly the source-free connected Gaussian, conditioning, wall, and Mayer construction of S4–S10. Keep its real cutoff sets fixed. In addition retain a fixed positive quadratic moment reserve after the unmarked comparison and failed-test estimates. The auxiliary exterior Gaussian is still integrated jointly, so its source-free cancellation remains in force.

Let O_Y(B,z) be a distinguished analytic carrying function on the actual background and wall domains, with complete carrier Y, and assume

$$|O_Y(B,z)| \le m_Y(1+E_Y(B))^{d/2}, \qquad E_Y(B)=\sum_{b\in Y^+}|B_b|^2, \qquad d<\infty. \tag{MT8}$$

Exponential weights on B can replace the finite enlarged carrier. Their column sum must be bounded in the same single energy. The coefficient norm of m_Y is the original rooted carrier norm with spare locality. A fixed power of carrier size is allowed and is paid from that spare locality.

Let L_V(O) be the coefficient of η in the connected output constructed from the finite source factor $1+\eta$ O, η^2=0. Include the deterministic zero-fluctuation readout and every local normalizing subtraction belonging to the actual construction. There is no source-dependent replacement of a frozen denominator. Then

$$\|L_V(O)\|_\kappa \le C_d\|m\|_\mu. \tag{MT9}$$

This is linear in the size of the mark and does not require it to be a small ordinary potential.

Proof. In a selected term of S9 insert the single O factor before the joint positive Gaussian comparison. For any retained reserve δ>0,

$$(1+E)^{d/2}e^{-\delta E} \le C_{d,\delta}. \tag{MT10}$$

Spend that fraction of the one quadratic reserve. All unmarked potential factors, wall-Cauchy factors, cutoff restrictions, and conditioning cancellations are unchanged. S10's covering argument is now rooted at Y and bounds the connected output carrier. The connected series with one distinguished vertex has the same convergent tree sum as the derivative series in S7. This proves MT9. No integral of a marked island is divided by its small mass.

The same proof with two specified marks O and R gives

$$\|L_V^{(2)}(O,R)\|_\kappa \le C_{d,e}\|m\|_\mu\|n\|_\mu \tag{MT11}$$

if $|R_Z|$<=$n_Z(1+E_Z)$ˆ{e/2}. This is a connected, jointly integrated estimate; it is not a product of two Gaussian or rare masses. For k fixed marks use prod_i$(1+E_i)^{\{\text{d_i/2}\}<=(1+\text{E_union})}${sum_i d_i/2}. Together with the finite partition formula, this gives the finite source-jet factor C_D^k(k!)ˆ{max(1,D/2)}.

AI44a–e deals explicitly with any bounded reference component quotient. The denominator and every cutoff boundary term stay exactly where the physical construction puts them.

There is a useful strengthened first-difference consequence. Suppose V is in the small QUADRATIC-growth formation ball, but its actual vertices also obey the polynomial estimate

$$|V_Y| \le g\, v_Y (1+E_Y)^{e/2}, \qquad \|v\|_\mu \le C. \tag{MT12}$$

The quadratic-growth norm may be O(g times a cutoff polynomial); it only needs to be below the fixed formation allowance. Marking one V vertex in the exact interpolation tV, 0<=t<=1, and using MT11 proves

$$\|L_V(O) - L_0(O)\|_\kappa \le C_{d,e} g \|m\|_\mu. \tag{MT13}$$

Thus the factor from the expanding canonical cutoff need not be paid in the one-step center estimate if MT12 is proved. All other unmarked vertices still use their ordinary small formation norm. The comparison between the fixed-cutoff L_0 and the full Gaussian has its separate polynomially weighted failed-test tail.

On a live or healed R component the identical proof uses S12 and S86: a fixed polynomial of its ONE still available energy consumes one fixed unused fraction of that energy. The output has the one rare majorant of the whole component, not its square. Engineering, promotion, and age factors are still retained. This lemma does not prove an incoming estimate such as MT8 for a source whose actual readout has not been controlled.

100.5 5. The Gaussian H center: the alias issue can be settled

Rescale the retained length to one, write a=1/n, and use the actual flat centered box-and-line Q. The gauge-invariant curvature of its curl-energy minimizer is that of $H_0\ P_C$, where

$$H_0 = G_a Q^* (Q G_a Q^*)^{-1}. \tag{MT14}$$

Use midpoint phases and coarse momentum θ near zero. Write q_{μ,ν}(θ) for the component-μ alias filter, ν=0 for the principal alias. The actual formula FE4a gives

$$q_{\mu,0} = 1 + O(|\theta|^2), \qquad q_{\mu,\nu}(0) = 0 \quad (\nu \neq 0), \qquad \lambda_0 = |\theta|^2 + O(|\theta|^4). \tag{MT15}$$

Let D_mu=|q_mu,0|^2+lambda_0 sum_{ν!=0}|q_mu,ν|^2/lambda_nu. Then the principal H_0 alias is bar(q_mu,0)/D_mu=1+O(θ^2), whereas every nonprincipal potential alias is

$$h_{\mu,\nu} = \frac{\lambda_0}{D_\mu} \frac{\overline{q}_{\mu,\nu}}{\lambda_\nu}. \tag{MT16}$$

It starts in degree three, not degree two. Multiplying by one or two fine differences does not lower its degree in theta.

One can make the needed second-difference conclusion uniform in n, without claiming an L-infinity bound for all second derivatives. Parseval for the finite averaging filter gives

$$\sum_{\nu\neq 0} |q_{\mu,\nu}(\theta)|^2 \le C|\theta|^2. \tag{MT17}$$

For an explicit majorant, FW9 for the actual filter gives |partial_theta q_{μ,ν}|<=C prod_i(1+|ν_i|)^{-1} on the principal small-momentum patch; its square is summable in four dimensions. Integrate this l2 derivative bound from zero, where all nonprincipal aliases vanish. The equivalent Parseval proof uses the actual finite offset filter and one bounded physical-position factor in its derivative. A bounded row sum alone, without this derivative control, would not suffice.

The multiplier of any two fine differences is bounded in modulus by C lambda_nu. Combining MT16–17, and using boundedness of P_C, proves

$$\left(\sum_{\nu\neq 0} |D_{a,i}D_{a,j}(H_0P_CB)_\nu|^2\right)^{1/2} \le C|\theta|^3|B|. \tag{MT18}$$

This is the normalized L2 norm in a fine offset cell. It is exactly the norm required for the quadratic directional-current energy. No sum of absolute second-difference alias amplitudes is used.

For the principal alias, the coarse gradient killed by the curl is i θ+O(θ^3), and the actual weighted Hodge projection is the ordinary transverse projection plus O(θ^2). The principal directional-current multiplier therefore has degree-two part

$$J^{(2)}_{\nu;\mu}(\theta)B = (-i\theta_\mu)(i\theta_\mu B_\nu - i\theta_\nu B_\mu). \tag{MT19}$$

The same formula follows directly from $d_r\ P_C{=}d_r$ and the principal expansion, so no singular longitudinal component has been discarded. The nonprincipal part MT18 contributes only degree six or higher to the quadratic energy; its cross term with the principal alias vanishes on a complete fine cell. Even without that orthogonality it starts in total degree five and cannot change the fourth potential jet.

For the pointed statement, retain TWO independent momenta. After absorbing the harmless fine-offset phase, write the directional-current leg as J_xi(θ)=M_2(θ)+R_xi(θ), where ||R(θ)||_{l2(xi)}<=C|θ|^3 by MT18–19. Cauchy–Schwarz in the offset cell bounds the difference of the bilinear legs at θ and phi by

$$C\{|\theta|^3|\varphi|^2 + |\theta|^2|\varphi|^3 + |\theta|^3|\varphi|^3\}|B|\,|B'|. \tag{MT19a}$$

Its total potential degree is at least five. This proves the pointed fourth jet directly; no diagonal polarization assumption about a two-momentum kernel is made. Consequently the degree-four potential symbol of sum_mu|J_{ν;μ}-J_nu/3|^2 is precisely the coarse H symbol. There is no additional dimension-six alias or interpolation contribution to its leading scalar quotient. Vacuum Wick contractions are still subtracted and recorded in the lower state.

For C, each linear curvature leg has its exact first jet by FE5–7; therefore its leading three-curvature symbol is unchanged. Gaussian contraction cannot turn a cubic into a quadratic. Any scalar linear contraction is zero by color invariance and is nevertheless harmless in the retained lower/contact bookkeeping. Thus, with the actual engineering factors restored, the free two-dimensional quotient transfer is

$$T_0 = L^{-2} I_2. \tag{MT20}$$

This is a calculation for the actual flat reconstruction, not a consequence of DN's nondegenerate free Gram matrix. The coupling normalization ratio in MT1 adds only I+O(g^2).

100.6 6. One precisely chosen primitive repair, and two different norms

Use the completed original slot PA1, $\Phi_j(U, J, B)$, whose physical value is the actual level-j minimum reconstructed from the current retained variables, including B. The near-window slot is the completed INNER minimum, not its arbitrary outer interpolated potential. Keep the actual hierarchy, action graph, inward conventions, and fixed boundary data of that slot. Write R_j for MC on this very determining problem.

We choose PA4, rather than leaving several candidate extensions interchangeable. With independent source-wall families s and t, define

$$\begin{aligned} \widehat{U}_{\mathbf{s}} &= \mathcal{R}_{j+1,\mathbf{s}}(U), \\ \widehat{J}_{\mathbf{s}} &= J_{\mathrm{phys}}(\widehat{U}_{\mathbf{s}}), \\ \Psi_{\mathbf{s}}(U, B) &= \Phi_j(\widehat{U}_{\mathbf{s}}, \widehat{J}_{\mathbf{s}}, B), \\ \mathcal{O}^{\mathrm{ext}}_{\mathbf{s},\mathbf{t}}(U, J, B) &= \mathcal{O}_j(\mathcal{R}_{j,\mathbf{t}}(\Psi_{\mathbf{s}}(U, B)))\,. \end{aligned} \tag{MT21}$$

In the last expression the independent argument J is ignored; the source slot uses the ACTUAL current of its repaired background. This matters when invoking the original full physical response theorem. On a physical outer minimum, at all source walls one, the inner repair is the identity, the current is the original physical current, Φ is the original completed inner minimum, and the outer repair is again the identity. Equality with the original insertion is therefore pointwise in B. This is PA5.

Only this primitive source extension changes. The action, real cutoffs, Gaussian kernels, coordinates, coarea factors, frozen reference denominators, and stored old-variable slots do not. An own-spacing noncritical stored link, or an uncompleted Haar factor in a live component, is not a slot of type MT21.

The input admission is concrete. The original own-scale C^1 relative bound gives the adapted $W^{1,p}$ condition in MC2; covariant plaquette telescoping gives its separate curvature bound. The inner MC output has the protected current and the low regularity required by the original completed response. Its full nonlinear response bound, including the near-window inner completion where required, puts Ψ in the outer MC low tube. Shrink the original core and physical fluctuation ceiling once by the fixed composition constants. There are two repairs per primitive, not a number increasing with the age of an activity. Use the same global determining problem and protect the entire primitive carrier with its complete collars. A separate artificial periodization would not give the exact endpoint identity and is not the construction in MT21. A primitive path leaving a regular domain stays in its original boundary/live state.

Here is the useful CANONICAL moment argument. Freeze one complete current carrier and its real canonical cutoff. Put every physical small input equal to t times its canonical input. Include both repairs and the physical current in MT21 when forming its t-family. The original response bound and the MC bounds, on their common smaller domains, give

$$\|\Psi_{\mathbf{s},t} - 1\|_{\mathrm{MC,input}} \le C|t|q_\infty, \qquad q_\infty = 1 + \|z\| + \max_{b \in Y^+} |B_b|, \qquad |t|q_\infty < c_0. \tag{MT22}$$

The input norm includes the first-order relative coordinates and curvature needed by the outer MC; $\Psi_{s,0}$=1. Here z includes the fixed canonical external data, with its actual first-order norm. Every determining, collar, exterior, and pinned input read by this coefficient is at the identity at t=0; if it is external data, it is scaled in this family as well. Holding a nonzero background fixed at t=0 would invalidate $\Psi_{s,0}$=1 and gives only MT25a. The physical cutoff bounds g times the coordinate SUPREMUM, and therefore puts t=g in a smaller disc. It does not imply that g times the l2 norm on an arbitrarily large carrier is small. The sup-norm analytic disc is established first. Only after Cauchy will q_∞ be bounded by a Gaussian energy polynomial.

MC and holomorphy, with the same protected collar, give for the final repaired field, denoted W(t),

$$\|F(W(t))\|_{H^{1+s}} + \|F(W(t))\|_\infty \le C|t|q_\infty. \tag{MT23}$$

One way to get the linear factor is to apply the uniform MC bound on the t disc of radius c_0/q_infinity and use its zero value at t=0. Banach-valued Cauchy on its fixed smaller concentric disc gives the displayed bound. This uses a genuine field/coupling disc, not a source disc for exp(z O).

The flux map is analytic with uniformly bounded derivative on this small plaquette branch. Its covariant divided-difference identity bounds D Φ(F) in L2 by C||D F||_2. Thus the actual directional-current-square insertion is bounded by C t^2 q_∞^2. For C-sharp the exact cube and transport formulas in AI32b imply the complex small-chart estimate C||F||_infinity||F||_2^2: each relative plaquette defect is O(a^2 F), the cube defect starts quadratically, and the single a^{-1} difference in AI32b leaves three curvature factors. All microscopic transports are bounded on this same chart. Therefore

$$|H_a(f;W(t))| \le C_f|t|^2 q_\infty^2, \qquad |C_a^\sharp(f;W(t))| \le C_f|t|^3 q_\infty^3. \tag{MT24}$$

The test norm and the explicit own-scale volume/engineering factors belong in C_f; there is no microscopic a^{-2} loss.

Let P_2 and P_3 denote the coefficients of t^2 and t^3 respectively. Divide MT24 by t^d, remove the removable singularity at zero, and use Cauchy once more on radius c_0/q_infinity. This proves

$$\begin{aligned} |g^{-2}H_a(W(g)) - P_2(z,B)| &\le Cgq_\infty^3, \\ |g^{-3}C_a^\sharp(W(g)) - P_3(z,B)| &\le Cgq_\infty^4. \end{aligned} \tag{MT25}$$

P_d is a degree-d canonical polynomial in the linearization of the CHOSEN repaired readout, with finite Wick/lower corrections kept separately. For arbitrary independent fine data that linearization includes MC's critical projection. To identify it with MT20, restrict the external data to the actual physical canonical coarse parametrization. Its tangent is the actual linear critical reconstruction; both repairs are the identity on that tangent graph. It is on this parametrization, after the physical own-current substitution, that the center is extracted.

On a fixed canonical external ball, q_∞^m is bounded by C_m(1+sum_{Y+}$|B_b|^{2)}${m/2}. Thus MT24 supplies MT8 and MT25 supplies its genuinely O(g) nonlinear difference, with one quadratic Gaussian reserve. A mere fixed-core bound g^{-2} times a constant would not prove MT25. The t-dependent radius and the vanishing field jets are essential.

There are TWO norms here. On the whole fixed ORIGINAL physical birth core of radius α, the corresponding estimate is only

$$|g^{-d}\mathcal{O}_d^{\mathrm{ext}}(U,J,gB)| \le C_d g^{-d}\left(\alpha + g\sqrt{E_{Y^+}(B)}\right)^d \le C_d(1+\alpha/g)^d(1+E_{Y^+}(B))^{d/2}, \qquad d=2,3. \tag{MT25a}$$

This follows by the same low-range and MC estimate with the actual external amplitude α, rather than g times a bounded canonical amplitude. It gives an analytic birth norm polynomial in g^{-1}, which MT9 permits for a finite mark. No smallness of that whole-core marked norm is asserted. Even a fixed smooth critical background with H nonzero has g^{-2}H growing as g^{-2}; the O(1) canonical bound cannot hold on that whole physical core.

For a source-wall coefficient with carrier X containing N_X current canonical coordinates, one may use E_X=N_X^{-1}sum_X|B_b|2. It has weights summing to one and q_∞^m<=C_m(1+||z||)^m $N_X^{m/2}$(1+E_X)^{m/2}. The fixed power $N_X^{m/2}$ is paid by spare source-carrier decay. There are only a fixed number of current coordinates per current cell; microscopic prefix bonds are read sets, not a second copy of the canonical Gaussian coordinates. This establishes the needed moment bound for each completed source carrier without an invalid large-carrier analytic-radius assertion.

Inherited source activities use MT25a and the original prepared core inclusion. Canonical Cauchy for the finite center uses MT24–25 on its separate smaller physical graph. Carrying these two bounds is legitimate; identifying them or claiming that one implies a coupling-uniform full-core norm is not.

On an own-spacing noncritical stored link variable one does not apply MC. Use the globally smooth compact H or C-sharp and its AI42/AI32c cold moment bound on its own record, with all engineering factors. Those records use the current physical spacing, not an arbitrarily long fine reconstruction treated as a single cell.

100.7 7. The one-step deduction and its separate history requirement

For the completed clean minimum slots specified in MT21 and PA1, MT22 holds on the common smaller original core and frozen real fluctuation cutoff. Section 8 below verifies MT12 for the actual clean unmarked vertices, in addition to their usual small quadratic formation bound. MT9, MT13, and MT25 therefore prove a rooted O(g) difference from the full Gaussian first-insertion output, with the separate rare and failed-test terms. Cauchy in a FIXED canonical external-field ball bounds all jets needed for the dimension-six projection. No g-dependent Cauchy loss is introduced after the normalization in MT1.

The C-even cubic extraction requires a correction to the C-odd argument. First extract the complete quadratic curvature kernel and choose its fixed gauge-covariant compact completion with the actual path/anchor convention. Subtract that entire completion. The remaining ordinary jets zero, one, and two then vanish; only now does its third jet factor through three linear curls. This procedure retains the nonlinear cubic terms forced by the quadratic Ward identity. Applying a three-curl homotopy directly to the original third jet would omit them.

Keep the complete quadratic tensor and its second path moments, the full even cubic jet, and all descendants and contacts before taking the scalar quotient. CD22–26 then supplies the strict regular-age factor r^s for the remainder, provided the resulting completed kernels have the stated common carrier norm and are read on the protected physical graph. The quartic-and-higher field remainder starts at r^8 and has r^2 after the same root and engineering count. The local finite projection itself only costs fixed Cauchy and carrier-moment constants.

If the same estimates also hold for the incoming marked stable/live state with its declared graded

norm, the full successive step has

$$T_j = L^{-2}\{I + E_j\}, \qquad \|E_j\| \le C g_j + C\,\mathrm{tail}_j + C\,\mathrm{rare}_j, \tag{MT26}$$

on the coupled quotient, together with the strict stable remainder. DN14 then supplies the ordered, possibly noncommuting center normalization. A finite two-dimensional quotient or a positive free Gram matrix alone does not supply MT26.

The remaining work is the marked-state requirement in the preceding sentence, not the already proved primitive moment calculation. MT21 and PA specify the near-window inner minima, physical current, and protected collars. MT9 and MT25a give the first completed activity on its original paired birth core with its explicit coupling factors. Later basic evaluations can therefore use the original S39 core inclusion, without repeatedly inserting MC into the whole old readout chain. A strict marked age norm, rather than that basic admission alone, is needed for MT36.

The complete marked Wilson induction, arbitrary operator mixing, and a continuum Yang–Mills construction remain unproved.

100.8 8. The actual clean unmarked input can be checked

This section verifies the clean MT12 input, instead of assuming that a small quadratic-growth norm automatically contains a factor g.

100.8.1 8.1 Fresh action and coordinate terms

The order of the original operations matters. I (2.10)–(2.13) removes the linear current pairing on the physical constrained critical graph and includes its quadratic correction in the Gaussian precision. I (3.1) then defines the independent-(U,J) extension from that already-subtracted expression. There is no assertion that an arbitrary independent J annihilates H_1 B. The off-shell formula simply no longer contains that negative power term.

The remaining action and gauge terms start at third order in the physical fluctuation B'. The constraint correction D(B') starts at second order. On the original smaller physical sup ball, analyticity gives, for example,

$$g^{-2}|V(H_1 B')| \le C g\, q_Y(B)^3, \qquad g^{-2}|(H_1 g CB, \Delta_1 H_1 h D(gCB))| \le C g\, q_Y(B)^3, \tag{MT27}$$

where B'=gCB-hD(gCB), and q_Y is a local sup or exponential response envelope. The quadratic D contribution is O(g^2 q_Y^4), and the coordinate Jacobian and Haar terms are bounded by a fixed g times polynomial envelope as well. There is no need to replace the physical bound g$\|B\|_\infty$ by its worst cutoff value in the single distinguished vertex. The other unmarked vertices still use the original small quadratic-growth norm.

The actual localization argument in II (1.37)–(1.40) bounds the action vertex cubically in B' before its g^{-2} factor is cancelled. Retaining that cubic bound gives MT27 with its completed carrier exponential. The same finite-coordinate analytic argument applies to the other fresh terms. Thus the fresh family obeys MT12 with a fixed polynomial degree and a rooted norm independent of the fine cutoff.

100.8.2 8.2 The inherited family: use its actual aggregated coefficient

There is a direct primary estimate, avoiding a potentially invalid interpolation between two different decompositions. II Lemma 1, (1.33)–(1.36), already sums the old ages and gives the actual localized inherited fluctuation coefficient V'*k(Y). On the original paired core it is analytic for* //B//{infinity,Y}<epsilon_1/g_k and obeys

$$|V_k'(Y;U,J,B)| \le CE_0\epsilon_1 e^{-\kappa' d(Y)}. \tag{MT28}$$

Most importantly,

$$V_k'(Y;U,J,0) = 0 \tag{MT29}$$

before the wall expansion is estimated. Indeed $D_s(0)$=$A_{0,s}(0)$=0 by II (1.14),(1.16), whence $H_{k,s}(0)$=0 by (1.17); also B'(0)=0 by (1.19). The original local fluctuation terms retain either their distinguished t_D derivative with argument (t zeta+t_D zeta_D)$H_{k,s}$(B'), or their distinguished relative response factor. At B=0 these are zero for EVERY wall value. The Ward rewrites and their paired marginal differences retain this factor. Linear wall differentiation and carrier filing therefore preserve MT29 coefficientwise. This is an identity of the actual extension, not a claimed off-shell equality between an arbitrary field and its reconstructed minimum.

Apply scalar Schwarz to z maps to V'*k(Y;U,J,zB) on the disc /z/<epsilon_1/(g_k//B//{infinity,Y}). On a fixed smaller physical cutoff, MT28–29 give

$$|V_k'(Y;U,J,B)| \le CE_0 g_k \, \|B\|_{\infty,Y} e^{-\kappa' d(Y)}. \tag{MT30}$$

The improvement is also explicitly allowed after II (1.29). The passage from the sup envelope to a Gaussian energy polynomial is made AFTER Schwarz and costs at most a fixed carrier power. Since II Lemma 1 has already performed the old age sum, no unjustified sum of constant birth bounds or of g_j^2 has been introduced. MT30 supplies the inherited part of MT12 on every admitted original paired background.

100.8.3 8.3 Compare the current-unit Gaussian density directly

For the center comparison put the external field and independent current on their fixed canonical ball, so their physical sizes are O(g). The actual current-unit precision Q_t, along the straight parameter path from the flat input to this input, has the original analytic row/column localization bounds. Its derivative has the same bounds with a factor g. Inverses are uniform on this common core. Put C_t=Q_t^{-1}.

Use the density in the SAME real canonical coordinates B and the SAME frozen reference cutoff. Its exact derivative is multiplication by

$$R_t(B) = \frac{1}{2}\mathrm{Tr}(C_t\dot{Q}_t) - \frac{1}{2}(B,\dot{Q}_t B). \tag{MT31}$$

For a completed carrier the trace counts only a fixed number of current-unit coordinates per cell. The trace part is O(g) times that cell count, and the quadratic part is O(g) times its local energy envelope. The operator products retain their completed carriers. This gives MT12 for this additional

comparison mark, including its constant/vacuum part. MT11 bounds the connected derivative of a marked coefficient; integrate t from zero to one.

This uses no Gaussian integration by parts, so no hidden derivative of a sharp cutoff is discarded. If the original presentation uses a background-dependent test, its exact cutoff-comparison tail must first be retained by the original tailflag construction. One cannot freeze a different test while claiming equality of the physical integrals.

100.8.4 8.4 A clean one-step result, with its scope

For the primitive minimum slots PA1, use the precise extension MT21 and its common protected source collars. Sections 6 and PA establish its admission, endpoint equality, full-core polynomial norm, and canonical polynomial moment. Sections 8.1–8.3 verify the unmarked O(g) input on the actual clean formation. The joint reserve in Section 4 now proves:

$$\left\| \mathcal{L}_j^{\mathrm{cl}}(\mathcal{O}_{d,g_j}^{\mathrm{ext}}) - \mathcal{L}_j^{\mathrm{Gauss}}(P_d) \right\|_{\text{canonical, rooted}} \le C g_j + C \,\mathrm{tail}_j, \qquad d = 2, 3. \tag{MT32}$$

The norm includes the fixed finite canonical external jets and a slightly weakened carrier exponential. It is uniform in microscopic cutoff and clean depth. The whole original-core output instead retains the polynomial bound MT25a. No claim that the two norms coincide is made.

Take the fixed dimension-six jet projection AFTER the complete marked coefficient has been formed, including all lower, descendant, and exact contact packets. The projection has bounded norm on the finite jets with their carrier moments. Here is the elementary bound that is used, rather than inferred from the dimension of the quotient. On a carrier with N current coordinates, separate-coordinate Cauchy bounds each coefficient of ordinary field orders two and three. Summing costs at most C N^3. Spatial derivatives of the kernel symbol through the required total momentum degree cost at most a fixed power of the carrier diameter. Both costs are absorbed by a fixed locality loss. Subtract the full quadratic covariant completion before applying the cubic Ward factorization, as in Section 7. The remaining reflection-even scalar quadratic symbol is the two-parameter symbol AI35a. In its conventions the H coefficient is F-E: it vanishes on the EOM direction E=F and equals one on H, for which (E,F)=(-2/3,1/3). The cubic coefficient is fixed by one nonzero color-triangle evaluation, as in DN7–8. Pointed descendants and the other lower coefficients are retained as separate coordinates. This defines a bounded finite-jet projection with fixed completions. Its canonical compact completion corrections are analytic in g and O(g); it does not presuppose a normalized interacting cubic. MT20 and MT32 imply the actual clean PRIMITIVE-column result

$$T_j^{\mathrm{cl}} = L^{-2}(I_2 + E_j^{\mathrm{cl}}), \qquad \|E_j^{\mathrm{cl}}\| \le C g_j + C \,\mathrm{tail}_j. \tag{MT33}$$

This settles the one-step principal matrix for these two primitive columns. It is stronger than a conditional claim that all MT12 inputs remain open. It does not, by itself, identify T_j^{cl} with the successive fundamental-matrix ratio after all old marked stable and live memory has fed back. That latter assertion requires the marked-state estimate specified next.

100.12 12. A proper fluctuation difference has a full-core small factor

There is a useful exact way to obtain a small full-core new-birth norm without falsely inferring it from MT38. Use ONE declared source readout $\Psi_{\mathbf{s}}(U,b)$, where b is the PHYSICAL fluctuation and s contains all its source-wall variables. On a completed carrier suppose this map is analytic for $\|b\|_\infty < \delta$, uniformly on the smaller physical input core and common wall polydisc. Both terms below use the same determining problem, the same walls, and the same background coordinates. Define

$$D_{j,k,\mathbf{s}}(U,b) := R_j(\Psi_{\mathbf{s}}(U,b)) - R_j(\Psi_{\mathbf{s}}(U,0)). \tag{MT46}$$

This is an exact identity inside the marked source factor. The second term is retained as the OLD deterministic tag; it is not declared a new activity and is not discarded. No unmarked integrand or denominator has changed.

Suppose the two readouts in MT46 have the same completed regular strict-age bound on that physical b disc:

$$(g_k/\alpha)^d \|R_j(\Psi_{\mathbf{s}}(U,b))\|_X \le C A_j (g_k/g_j)^d (r^s + r^\beta + r^2) e^{-\mu \mathcal{D}(X)}. \tag{MT47}$$

Here α and δ are fixed physical radii; replacing α by $\alpha+\delta$ changes only the fixed constant. For the direct protected readout MT47 is MT42–43 before its canonical energy substitution. Its validity must be checked on the entire b disc, not just on the real cutoff.

The zero $D_{j,k,\mathbf{s}}(U,0) = 0$ holds for EVERY complex wall value. Every wall localization is linear in the marked carrying function, so it preserves this zero before the connected formation. Apply one-variable Schwarz to $z \mapsto D(U, zB/\|B\|_\infty)$ on $|z| < \delta$. At the physical value b=g_k B, with $g_k\|B\|_\infty \le \delta/2$, this proves

$$\begin{aligned}(g_k/\alpha)^d \|D_{j,k}(U, g_k B)\|_X \le C A_j \, \frac{g_k}{\delta} \|B\|_\infty \, (g_k/g_j)^d \\ \times (r^s + r^\beta + r^2) e^{-\mu' \mathcal{D}(X)}.\end{aligned} \tag{MT48}$$

The fixed loss μ-μ' pays the wall-Cauchy and carrier polynomials. The analytic evaluation uses the sup norm; only afterward is $\|B\|_\infty$ bounded by the single local Gaussian energy. The finite-mark lemma therefore gives the SAME g_k/delta factor in the full-core norm of a completed connected new birth. This applies also to the quartic analytic tail. No O(α) estimate for that tail on its own α core is asserted.

The corresponding canonical center estimate uses the graded table MT44, the O(g_k) nonlinear readout and unmarked-vertex comparisons, and the exact free lower filing below. In particular it does not try to prove smallness by taking absolute values of a free quadratic Gaussian contraction before recording its vacuum term.

For a completed universal regular family this is a specific proper-difference lemma, not a proposed change of measure. To turn it into an all-age source recursion, one still has to identify the old

deterministic term with the chosen next prepared source readout, on the same source-wall carriers. Equality at the physical endpoint alone is insufficient for that off-shell identification. A shared protected-field/direct-core readout for old ages and the original prepared readout for finitely many recent ages must be declared and localized as one source construction before MT48 is used for that recursion.

100.13 13. The exact free packet and its lower rows

The free reference in the proposed interaction-picture argument need not be an invented independent Gaussian. It has the existing exact nested-projection realization of Section 10.
Work on the positive flat gauge quotient, with harmonic zero modes retained separately. Let H_0 be the fine Maxwell precision, $Q^{(j)}$ the actual composed boxed coarsening, and set

$$H_j = (Q^{(j)} H_0^{-1} Q^{(j)\mathrm{t}})^{-1}, \qquad P_j = H_0^{-1/2} Q^{(j)\mathrm{t}} H_j Q^{(j)} H_0^{-1/2}. \tag{MT49}$$

These are precisely the Schur precisions and nested orthogonal projections, with $P_i P_j = P_{\max(i,j)}$. The running quadratic counterterms give the one fine precision

$$K_n = H_0^{1/2} \left\{ \sum_{\ell=0}^{n-1} x_\ell (P_\ell - P_{\ell+1}) + x_n P_n \right\} H_0^{1/2}. \tag{MT50}$$

Every x_l is positive; no pointwise monotonicity is required. The projection differences are independent Gaussian channels. Conditioning successively on the nested Q data is therefore EXACTLY conditioning this one Gaussian in stages. Its mean is the classical nested reconstruction, independent of the scalar channel weights.

More explicitly, if $Q^{(k:j)}$ is the actual coarse map from level j to k, the conditional mean on level j is

$$H_{j:k} = H_j^{-1} Q^{(k:j)\mathrm{t}} H_k, \qquad H_{i:j} H_{j:k} = H_{i:k}. \tag{MT51}$$

The identity follows by cancelling the middle Schur precision and using Q^{(k:j)}$Q^{(j:i)}$=$Q^{(k:i)}$. This statement concerns the actual induced coarse precision, not a replacement by a nearest-neighbor coarse Maxwell action. The raw linear constraint differs by a gauge gradient, so these are the same physical curvature means on the quotient. The finite harmonic data are held fixed.

At first source order the normalized Gaussian source coefficient is simply conditional expectation. For a completed principal quadratic curvature packet, Wick's formula gives its value on H_{j:k}v plus a covariance contraction. That contraction is a vacuum/source-test coefficient and is retained in the lower row. For a principal invariant cubic packet, the possible single contraction is an invariant linear Lie-algebra tensor. It vanishes for a compact simple gauge algebra; the flat covariance is color diagonal. Thus, modulo those explicit lower rows,

$$\mathcal{G}_{j:k} E_2(v) = E_2(H_{j:k} v), \qquad \mathcal{G}_{j:k} E_3(v) = E_3(H_{j:k} v). \tag{MT52}$$

The covariance contributions are kept before any absolute bound or local normalizing subtraction. With several sources the corresponding coincidences belong to the exact joint contact module, not to independent one-mark copies. The present assertion is only the first marked row. A product of cutoff Gaussians is not silently identified with MT50: use the exact reference-cutoff tailflag comparison before invoking the untruncated Gaussian tower.

For the complete UNIVERSAL old-root family the free stable-to-center statement has a direct phase check. Let θ and φ be the two current external momenta and η the pointed test momentum, with $\theta+\varphi+\eta=0$. In an old-root subcell $x=q/n$, the alias phase is

$$\frac{1}{n^4}\sum_{q\in\{0,\dots,n-1\}^4} e^{2\pi i(\nu+\mu)\cdot q/n} = \mathbf{1}_{\nu+\mu=0 \bmod n}. \tag{MT53}$$

A fixed source anchor contributes $e^{i\eta z}$; it does not change this root phase. Hence a principal and a nonprincipal alias cannot contribute together. For the induced level-j precision this assertion is not obtained by substituting a nearest-neighbor coarse Laplacian into MT16. Instead use the exact identity $H_{j:k}=Q^{(j)}H_{0:k}$ and the flat cochain identity $d_jQ^{(j)}=Q_2^{(j)}d_0$. A nonprincipal old-coarse alias receives only nonprincipal fine aliases. The whole Q_2 tail is an L2 contraction, so MT15–18 gives curvature order at least three, uniformly even when the microscopic prefix is long. Two such nonprincipal curvature legs therefore have total potential degree at least six. Principal/principal terms have zero dimension-six jet when the lower quadratic family and the full degree-four potential moments have already been removed. A cubic nonprincipal leg likewise raises the minimum potential degree from three to at least five.

The exact dimension-four family must have its OWN free evolution retained. Merely requiring a dimension-six-zero coefficient while leaving a bare F-squared term in the stable state would be wrong: its principal interpolation correction can generate a degree-four potential term. Likewise MT53 is a statement after the COMPLETE old-root and subcell-phase sum. A localized smooth root partition is summed back to one before this free projection, or its explicitly differentiated test descendants are kept until that sum. No principal/nonprincipal cancellation is claimed for an individual root-dependent history member.

Together MT49–53 identify the exact free packet that can be carried with its original birth tag. MT48 identifies the proper fluctuation difference that can receive a new small birth. The remaining clean-recursion interface is the actual common source-wall/readout identification in Section 12: the hybrid must have one declared complete read set, an admitted completion on every required wall value, and the exact deterministic old-tag identity used by the next step. The physical own-current restriction must also remain compatible with the full lower-dimensional, current, EOM, and contact files under their subsequent joins. The elementary free lower rows in MT52 do not prove that complete bookkeeping theorem. Finally the live and R marked rows require their actual birth radii, healing and promotion rules, and once-owned rare reserve. None of these global statements follows merely from the primitive columns or the local graded formation estimates proved here.

102.1 1. Use the same variational problem before asserting transitivity

Fix one finite fine action graph, its fixed exterior and allowed gauge group, and a genuine nested determining hierarchy. Denote the actual holomorphic raw determining maps by C_j. The nesting required below is the literal identity

$$C_k = C_{k:j} \circ C_j, \qquad j < k. \tag{HI1}$$

On a periodic uniform hierarchy these are the actual composed raw means. On a mixed hierarchy HI1 includes the retained outer rows, the inward-junction convention, the fixed boundary rows, and every averaging read set. Changing the action graph or independently periodizing two windows is not covered.

Let S be the SAME fine Wilson functional. Let $U_j(\mathrm{v})$ be its unique admitted small critical orbit with $C_j(U_j(\mathrm{v}))=\mathrm{v}$. Define the retraction

$$P_j(U) := U_j(C_j U). \tag{HI2}$$

All statements are on a common complex domain on which these branches and the indicated compositions are admitted. Equalities of fields below may be understood modulo the prescribed gauge; every source under consideration is gauge invariant, including the covariant transformation of its physical current.

The actual source minima have this common-action property. In Bałaban I, (0.21) and (1.1) define U_k by the fine functional A(U) with M^kU=V. Equation (0.22) separates the running coefficient of $\mathrm{A}(U_k)$ from $E_k(U_k)$. The saddle used in (2.1)–(2.3) is that of the classical $\mathrm{A}(U_k(\mathrm{V}))$, with the gauge-fixing term selecting its representative. The inherited effective interaction does not redefine this background minimum.

Multiplying S by a nonzero scalar does not change P_j. Adding to the level-j objective a function of C_j alone also does not change its constrained critical points. An effective interaction which is not constant on the C_j fibers would be a different matter.

Complex retraction lemma. On the stated common domain,

$$P_j P_k \simeq P_k P_j \simeq P_k, \qquad j < k. \tag{HI3}$$

For the second composition, exact target preservation and HI1 give $C_k(P_j U) = C_k U$. The unique level-k branch therefore gives $P_k(P_j U) \simeq P_k U$.

For the first composition, $W = P_k U$ is critical for S subject to C_k. Every infinitesimal variation preserving C_j also preserves C_k, because of HI1. Hence W is critical subject to $C_j = C_j W$. Equivalently, if $dS(W) = DC_k(W)^{\mathsf{t}}\lambda$, then

$$dS(W) = DC_j(W)^{\mathsf{t}} DC_{k:j}(C_j W)^{\mathsf{t}}\lambda.$$

The transpose is the invariant bilinear transpose in the actual physical pairings. Uniqueness of the admitted level-j branch proves $P_j W \simeq W$.

This proof is a complex KKT argument, not an extension from a physical endpoint by assertion. It does not use positivity for complex fields. Existence and branch uniqueness, as well as domain admission, remain the actual classical inputs. It also makes clear why arbitrary critical points of different effective actions would not suffice.

102.2 2. Localized responses do not automatically inherit HI3

II (1.14), (1.16), and (1.17) give $D_s(0)=A_{0,s}(0)=H_{k,s}(0)=0$ for every complex wall value. This is the zero of the outer fluctuation response. It does not say that every inner reconstruction at nonzero determining data is a critical map of the same wall-dependent action. II (1.5)–(1.10) localizes the defining propagators by their random-walk factors; the analytic estimates use their bounds, not a newly proved common variational principle for all weakened maps.

Here is an elementary counterexample to the missing inference. For $x = (x_1, x_2, x_3)$, constraints $C_1x = (x_1, x_2)$, $C_2x = x_1$, and small fixed e, put

$$\begin{aligned} P_{1,s}x &= (x_1, x_2, e(1-s)x_1), \\ P_{2,t}x &= (x_1, e(1-t)x_1, 0). \end{aligned} \tag{HI4}$$

Each is analytic, uniformly bounded on a fixed wall disc, idempotent, and preserves its target. At walls one they are the genuine nested orthogonal retractions for the real quadratic action $S(x) = |x|^2/2$, with its holomorphic quadratic extension. For real walls near one, each is even a genuine constrained minimum: use respectively

$$\begin{aligned} S_{1,s}(x) &= \tfrac{1}{2}\{x_1^2 + x_2^2 + (x_3 - e(1-s)x_1)^2\}, \\ S_{2,t}(x) &= \tfrac{1}{2}\{x_1^2 + (x_2 - e(1-t)x_1)^2 + x_3^2\}. \end{aligned} \tag{HI5}$$

Nevertheless

$$P_{1,s}P_{2,t}x = (x_1, e(1-t)x_1, e(1-s)x_1),$$

which differs from $P_{2,t}x$. The even analytic observable x_3^2 detects the difference. Thus endpoint agreement, analyticity, target preservation, individual uniqueness, and small bounds do not prove a wallwise tower. The example is a counterexample to that inference, not a counterexample to an actual Wilson identity.

Literal zero-fluctuation compositions can always be kept as finite tags wherever their domains compose. Their algebra is then exact, but neither a direct age estimate nor a uniform domain follows merely by naming the composition. HI3 alone does not collapse such a word at nontrivial walls.

102.3 3. An actual global source identity on a complex domain

Let R_j be a gauge-invariant completed birth function on its original paired core. When using the coarse physical kernel representation, write its declared physical pullback as

$$R_j^\natural(v) := R_j(U_j(v), J_{\mathrm{phys}}(U_j(v))). \tag{HI6}$$

If this pullback has been localized, every resulting member retains the actual response read set and the physical-current substitution. The coarse kernel/anchor norm in MA20–24 is a suitable incoming class. It is not obtained by replacing independent-current jets with physical jets after estimating them.

Let $\Phi_k(W, J_{\mathrm{phys}}(W), b)$ be the ORIGINAL completed level-k MINIMUM READOUT, with its original physical fluctuation coordinate b, based at an admitted level-(k+1) critical field W. At its physical inputs and every admitted physical fluctuation it is the actual level-k minimum with the original determining data. Thus the later application of $R_j^\natural \circ C_j$ is the genuine nested critical pullback, not the projection of an arbitrary fine field. At $b = 0$ its determining data are those inherited from W. The actual response is therefore W up to gauge. This is also visible from I (2.3) and the zero relative response in II (1.17)–(1.19). Inner window completions must use their declared action graph and actual determining and boundary data; a different local saddle is not substituted.

For arbitrary complex U in the common admitted domain set W=P_{k+1}U and define the unweakened source

$$F_{j,k}(U, b) := R_j^\natural(C_j \Phi_k(P_{k+1}U, J_{\mathrm{phys}}(P_{k+1}U), b)) \,. \tag{HI7}$$

The current argument in this expression is physical. Its independent source-core extension is deliberately the one in HI7; it is not identified with an arbitrary original off-shell (U,J) formula. At the physical Wilson inputs the original completed response makes HI7 the same insertion pointwise in b. The unmarked kernel is untouched.

At zero fluctuation HI3 gives the stronger exact identity

$$F_{j,k}(U, 0) = R_j^\natural(C_j P_{k+1}U) = R_j(P_{k+1}U, J_{\mathrm{phys}}(P_{k+1}U)) =: T_{j,k+1}(U). \tag{HI8}$$

This holds for EVERY U in that complex domain, because P_{k+1}U is itself on the actual critical branch. It is not merely equality on real critical inputs. The physical current follows from that same field and gauge transformation.

The deterministic tag has a direct meaning at every later level: $T_{j,l}(U) = R_j^\natural(C_j P_l U)$. Any intervening UNWEAKENED retractions collapse by HI3. Its birth function R_j is retained; no sequence of repairs is composed into it.

For mixed or changing local windows, this conclusion requires one genuine common hierarchy before localization. Two windows with different prescribed boundaries cannot be identified by HI3. Their finite local inverse extensions may still occur as factors in an exact localization of the same global map.

102.4 4. Two wall families suffice for the exact source algebra

Choose an admitted localized source family $F_{j,k,s}$(U,b) with $F_{j,k,1}$=$F_{j,k}$. It may be the PA/MC pre-repair, original response, and post-repair construction, with every repair taken on the declared actual determining problem. Define

$$D_{j,k,\mathbf{s}}(U, b) := F_{j,k,\mathbf{s}}(U, b) - F_{j,k,\mathbf{s}}(U, 0). \tag{HI9}$$

Separately choose an admitted localization $T_{j,k+1,t}$(U) of the DIRECT tag HI8. Its source walls t may be independent of s. For example use one localized current-level retraction before the actual old coarsening and physical birth function. There is no requirement that $F_{j,k,s}$(U,0)=$T_{j,k+1,t}$(U) away from walls one. The global identity needed by the integral is

$$F_{j,k}(U,b) = T_{j,k+1,\mathbf{1}}(U) + D_{j,k,\mathbf{1}}(U,b). \tag{HI10}$$

By HI8 this identity holds on the full common complex input domain. Start from HI10 and localize its two terms. One must not first localize F_s and then silently discard the coefficientwise difference between F_s(U,0) and a different T_t.

Suppose the two actual expansions are absolutely convergent:

$$T_{\mathbf{1}} = \sum_X T_X, \qquad D_{\mathbf{1}} = \sum_Y D_Y, \qquad D_Y(U,0) = 0. \tag{HI11}$$

The last zero is coefficientwise because wall differentiation, evaluation, and FTC integration are linear in the SAME difference HI9. Each coefficient keeps its own complete carrier, including the protected collar, all weak inverse/vertex read sets, raw prefix paths, determining data, test offsets, and fixed exterior data that its definition actually reads.

For any common rooted exponential norm for which both expansions are controlled, their disjoint tagged sum obeys

$$\|(T_X)_X \sqcup (D_Y)_Y\|_\kappa \le \|T\|_\kappa + \|D\|_\kappa. \tag{HI12}$$

There is no product of two localization costs merely because the wall families differ. Each is evaluated on the same external field domain and their own complete sums recover HI10. If a later operation multiplies coefficients, it uses the union of their actual carriers and the corresponding joint norm.

For a square-zero scalar source η, the exact normalized first-mark identity at fixed external U is

$$\begin{aligned}\log \int K_U(B) e^{V_U(B)} [1 + \eta\{T(U) + D(U,B)\}]\, dB = {} & \log Z_0(U) \\ & + \eta T(U) + \eta \frac{\int K_U e^{V_U} D(U,B)\, dB}{Z_0(U)}.\end{aligned} \tag{HI13}$$

Here Z_0 is the original source-zero integral, and the formula is used only where that normalized connected branch is admitted. T is independent of the variables integrated in this formula. It passes deterministically; it is not repeatedly formed as a new interacting activity. Thus its size need not have a fresh coupling-small factor.

The corresponding unnormalized factorization is

$$\begin{aligned}&\int K_U(B) e^{V_U(B)} e^{T(\boldsymbol{z},U) + D(\boldsymbol{z},U,B)}\, dB \\ &\qquad = e^{T(\boldsymbol{z},U)} \int K_U(B) e^{V_U(B) + D(\boldsymbol{z},U,B)}\, dB.\end{aligned} \tag{HI13a}$$

It is valid in any finite commutative nilpotent source algebra when T includes ALL joint deterministic coefficients. It does not justify replacing joint contact coefficients by products of separate marks. Nor can a T which depends on old variables inside a later joint live integration be pulled out of that entire integration. It remains a multiplier in the retained old-variable operator. The separate finite-source pushforward/contact argument addresses that case.

102.5 5. Quantitative domain conditions for this scoped construction

The identities above are unconditional once all displayed functions are defined. Uniform estimates require a common domain; the following conditions specify it without an iterated radius argument.

Fix the actual original paired birth radii, a smaller MC input tube radius e_*, a fixed physical response disc δ, and fixed source-wall radii R_s,R_t>1. Let C_P,C_{Phi},C_out denote the already proved one-current-level range constants for the repaired background, original completed response, and repaired output. Choose the input radius e and δ ONCE so that

$$\begin{aligned} C_P e &< e_*/4, \\ C_{\mathrm{Phi}}(C_P e + \delta) &< e_*/4, \\ C_{\mathrm{out}} C_{\mathrm{Phi}}(C_P e + \delta) &< e_{\mathrm{reg}}. \end{aligned} \tag{HI14}$$

The last radius is the protected regular output size. It is chosen below the fixed clean prepared-core admission margin. There are a fixed number of maps in this composition, not one map per past age. The MC wall radii are those of the adaptive localization cells, with the per-box wall count of MB/MC; they do not shrink with microscopic depth. Original response walls have their own fixed admitted radius. Their product polydisc is legitimate after this one fixed range choice, with the following joint-domain qualification.

If action, coordinate, or propagator walls are already active in an input read by either source family, the constants and physical fluctuation disc in HI14 must hold on their JOINT domain with both source-wall families. Separate holomorphy in each family is not a proof of that product-domain bound. Alternatively one may perform the exact source expansion first, in the fixed original fluctuation coordinates, and retain its coefficient functions unchanged while subsequently localizing the unmarked kernel. In that order the later action walls do not become new arguments of the source readout. Any later wall-dependent coordinate substitution must still be checked against the same source fluctuation disc. The original real cutoff and the conditioned exterior remain where the actual construction puts them.

HI14 is a CLEAN fixed-core choice. For R births with shrinking α_j, γ_j, recent original-core admission may allow only a shrinking physical fluctuation radius δ_j. It does not then supply the fixed disc δ. The later direct conditions HI15 can admit an old R carrier on that fixed disc once its actual age is large enough, but the waiting age depends on its shrinking radii. Until that point a Schwarz factor is g_k/δ_j, with the original radius/promotion bank retained; it is not replaced by g_k/δ.

For a sufficiently old contractible completed carrier X, let r=L^{j-k} and D_X be its diameter in birth units. The direct old-core conditions from DA have the explicit form

$$r D_X < c_{\mathrm{chart}}, \qquad C_X r^2 e_{\mathrm{reg}} < \alpha_j/2, \qquad C_X r^3 e_{\mathrm{reg}} < \gamma_j/2. \tag{HI15}$$

The last two radii are the field and current radii of the actual birth core; neither is omitted for an R birth. C_X includes the fixed path, test, and collar geometry of this member. On this domain the actual raw coarsening of the protected field is evaluated in its straightened analysis gauge, as in CB, while the gauge-invariant source value remains the raw one.

The source collar protects the UNION of all read sets used by the two terms of HI9, not just the support of a final curvature polynomial. In the old direct case rD_X is bounded, so one fixed current-scale enlargement can contain this union. In other cases the source carrier records the needed enlarged set; a filled bounding box of unaccounted volume is not free.

For finitely many recent ages, or for carriers not eligible for HI15, the original prepared representation can be used on its original admitted paired core. The admission theorem is applied to the SAME bounded birth function with its actual original arguments. It is not inferred by composing a map with a generic bound C times the radius at every age. Its unweakened endpoint is compared with HI7 only after the input has been placed on P_{k+1}U, where the genuine nested identities apply. The intermediate wall values need not coincide.

This recent/prepared alternative supplies basic original-core admission. A sharp canonical graded bound for its wall values still requires its stated physical own-current representation, or a separate bound proving the same graded norm. Ordinary paired-core admission alone does not convert arbitrary independent-current jets into the physical kernels of HI6. Long, wrapping, or wild-crossing carriers likewise retain their actual original tail or live norm until that estimate has been paid.

For a known completed family with incoming coefficient bound A_X exp(-μ d(X)), fixed path/coordinate-count polynomials are absorbed by a spare locality exponent. New localization cells and collars must be charged in the same completed carrier norm. Neither HI3 nor HI10 proves such a coefficient bound from analyticity alone. This is the quantitative place at which a claimed global hybrid must be checked.

102.6 6. The proper-difference gain is compatible with the two-family split

Suppose the actual D_s family satisfies the MT47 full physical disc bound, with its complete regular root/phase packet and the joint-wall or ordered-expansion qualification just stated:

$$(g_k/\alpha)^d \sup_{\|b\|_\infty<\delta,\mathbf{s}} \|F_{j,k,\mathbf{s}}(U,b)\|_X \le N_{j,k,X}. \tag{HI16}$$

Since D_s(U,0)=0 for all walls, radial Schwarz and the same linear wall localization give

$$(g_k/\alpha)^d \|D_{j,k,X}(U,g_kB)\| \le C\frac{g_k}{\delta}\|B\|_\infty N_{j,k,X}. \tag{HI17}$$

If N carries the graded relative coupling ratios and strict age weight of MT43–44, those weights remain in HI17. The canonical supremum bound precedes the one-joint-energy substitution. The two-family split does not weaken this zero-fluctuation estimate: the independently localized deterministic T never enters it.

For a member admitted only on a physical disc δ_j, HI16–17 use that radius throughout. A fixed clean radius cannot be inferred from ordinary R-core admission.

Conversely, HI17 says nothing about the admission of a different readout substituted after proving HI16. That substitution is exactly the error the two-family definition avoids. It fixes F_s before taking its difference and fixes T_t separately from the direct global identity.

102.7 7. What has closed, and the remaining compatibility lemma

For the fixed genuine regular hierarchy described here, the following facts are proved:

1. The unweakened retractions have a tower identity on their common complex domain, for the actual common Wilson action.
2. The zero-fluctuation tag of the declared physical source extension is the direct current critical tag on that domain.

3. The proper difference and direct old tag may be localized with independent source-wall families. Exact source algebra does not demand a wallwise tower or coefficientwise equality between their off-wall zero readouts.
4. The old tag can pass deterministically, while only the proper difference uses the coupling-small formation bound.

Thus a missing wallwise identity $P_{j,s}P_{k,t} = P_{k,t}$ is not the right remaining requirement. It is neither proved nor needed by this construction.

The remaining uniform statement is a bounded source-preparation lemma. For every ACTUAL completed incoming regular birth one must choose the recent/prepared or old/direct member on one fixed common field domain so that:

- the complete read sets and boundaries are those of the same global determining problem, with all required protected collars included;
- both source localization families have depth-independent wall radii and per-carrier exponential bounds;
- the proper-difference family has HI16 on the full physical fluctuation disc, including the canonical graded norm and the physical own-current restriction;
- the newly formed physical pullback returns to the completed coarse kernel/anchor/phase class with its lower and contact coefficients still present.

The first two-family algebraic identities cannot supply the last invariant-class estimate. Basic original-core inclusion does not prove it either. The clean universal-family assumptions of MA20–24 and CB provide substantial regular inputs, but a history-dependent coefficient need not have their translation pairing. Live and R objects still require their joint operator norms, changing radii, healing/promotion rules, and rare reserve.

This note therefore removes one unnecessarily strong off-wall compatibility demand and proves an exact usable replacement. It does not claim an all-depth clean recurrence, a complete operator-mixing theorem, or a continuum Yang–Mills construction.

103.2 1. Finite Wilson observables and two different truncations

Let

$$\mathcal{M}_a = G^{E_a}, \qquad d\mu_a(U) = Z_a^{-1}\rho_a(U)\,dU, \qquad \rho_a(U) = e^{-S_a(U)} > 0, \tag{XP1}$$

where G is compact, the lattice is finite, and dU is product Haar measure. Boundary links, if any, are fixed data. For the Wilson measure $S_a = \beta_a S_0$.

The fine observable class includes scalar components of finite polynomials in link matrices, their inverses, compact plaquette fluxes, finitely many covariant lattice differences, and finite parallel transports. It is closed under multiplication and the invariant bond derivatives. Every such observable is smooth on the entire compact configuration space. Powers of a^{-1}, normalization matrices, and tested sums are permitted; their possible divergence as a tends to zero is a quantitative issue, not an obstruction to the identities here.

Tensor or colored observables can be carried componentwise in a finite tensor module. A scalar source couples to a specified component or invariant contraction. Products keep their original matrix order

before taking the stated contractions. No commutativity of matrix multiplication is used.

A polynomial in a local logarithm is not by itself a globally smooth Wilson observable. It can be used on its actual admitted chart with its original cutoff. Derivatives of that cutoff are then retained. The distinction matters in Sections 5–6.

For m distinct source labels use the squarefree algebra

$$\mathcal{J}_{[m]} = \mathbb{C}[\eta_1, \dots, \eta_m]/(\eta_1^2, \dots, \eta_m^2). \qquad \eta_I = \prod_{i \in I} \eta_i. \tag{XP2}$$

Repeated insertions of the same physical observable use distinct labels. The coefficient of $\eta_1 \cdots \eta_m$ in

$$\exp_{\mathcal{J}}\left(\sum_{i=1}^{m} \eta_i O_i\right) = \prod_{i=1}^{m} (1 + \eta_i O_i) \tag{XP3}$$

is their product. Alternatively one may use the quotient by all monomials of total degree greater than m, with the usual factorial convention. Everything below is coefficientwise and works for either finite algebra.

The nilpotent exponential and logarithm are finite sums. They need no ordinary exponential source radius for a cubic or higher observable. If $A = \sum_{\varnothing \neq I} \eta_I A_I$, then

$$[\eta_I] e^A = \sum_{\pi \in \Pi(I)} \prod_{B \in \pi} A_B, \tag{XP4}$$

where $\Pi(I)$ denotes the set of partitions of I.

This is a truncation in **source degree**. A later Taylor projection in the field variables U, J is a different operation. The latter generally discards a nonzero function unless its complement is retained.

103.3 2. The exact transport functor is a functor of densities

103.3.1 2.1 Kernel extension and composition

A state at a history step contains every current and still-retained old variable, together with its actual tests and geometric labels. A source-free linear operation can be written schematically as

$$(\mathcal{K}F)(y) = \int k(y, x) F(x)\, d\lambda(x). \tag{XP5}$$

The kernel may include fixed coordinate densities, constraints, source-zero potentials, and selectors. Delta constraints are interpreted as their actual disintegrations or as distributions. Positivity or normalization of each displayed operator is not needed for linearity.

Extend this operation to $\mathcal{J}_{[m]}$-valued states by

$$\mathcal{K}^{\mathcal{J}}\left(\sum_I \eta_I F_I\right) = \sum_I \eta_I \mathcal{K} F_I. \qquad [\eta_I] \mathcal{K}^{\mathcal{J}} F = \mathcal{K}[\eta_I] F. \tag{XP6}$$

Whenever the original iterated integrals exist, Fubini gives

$$\mathcal{K}_2^{\mathcal{J}} \mathcal{K}_1^{\mathcal{J}} = (\mathcal{K}_2 \mathcal{K}_1)^{\mathcal{J}}. \tag{XP7}$$

Thus every source coefficient follows the same sequence of kernels as the source-zero body. Finite source differentiation, source-jet truncation, a fixed change of variables, and an exact finite partition of unity commute with this extension.

For an infinite localized or connected representation, XP7 applies after coefficientwise absolute convergence has been established, or to its finite approximants with a separately justified limit. Finite nilpotence alone does not justify exchanging an unbounded carrier sum with an integral.

103.3.2 2.2 Products must be joined before integration

In general

$$\mathcal{K}(FG) \neq (\mathcal{K}F)(\mathcal{K}G). \tag{XP8}$$

For example, conditional integration of a centered nonzero Gaussian variable gives $\mathcal{K}B = 0$ and $\mathcal{K}B^2 > 0$. Two marks entering one still-live component therefore stay inside its one joint kernel. A product of their separately integrated masses is not its coefficient.

The exact product rule is instead XP4 inside XP5:

$$[\eta_I]\mathcal{K}(e^A) = \int k(y,x) \sum_{\pi\in\Pi(I)} \prod_{B\in\pi} A_B(x)\, d\lambda(x). \tag{XP9}$$

Literal conditional product kernels and product input states allow factorization. Overlapping live histories, or states that have not actually split, do not.

This also explains why an exact coarse state need not remain in the algebra of finite Wilson polynomials. Integration can generate a general analytic function, a retained integral kernel, or a density with cutoff boundaries. The source algebra is finite; the space of its coefficient functions need not be finite dimensional.

103.3.3 2.3 Coarse normalization is optional

If $\mathcal{K}\rho$ is positive, one may define a conditional normalized map by dividing by this body. That map still does not preserve products. Nothing in XP5–XP9 requires such a division.

At the terminal physical integral the scalar source-zero mass $Z_a > 0$ is available. If

$$Z(\eta) = Z_a \left(1 + \sum_{\varnothing\neq I} \eta_I m_I\right), \tag{XP10}$$

then the finite logarithm gives

$$[\eta_I] \log \frac{Z(\eta)}{Z_a} = \sum_{\pi\in\Pi(I)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_B. \tag{XP11}$$

This is the usual connected coefficient, with one original global source-zero normalization. It supplies no bound on a conditional island denominator.

103.4 3. Coarea, frozen cutoffs, and the matched reference move

103.4.1 3.1 A coordinate identity carries the entire mark

On a regular coordinate patch for a deterministic observation $q : \mathcal{M}_a \to \mathcal{N}$, a fiber chart $U = T_b(v, \xi)$ gives

$$q_*(\rho F)(v) = \sum_b \int \chi_b(T_b(v,\xi))\rho(T_b(v,\xi))F(T_b(v,\xi))J_b(v,\xi)\, d\xi. \tag{XP12}$$

Here the chart partition, Jacobian, allowed domain, and any gauge quotient are the actual ones. This identity holds for each finite source coefficient of F.

No new source-dependent coordinate change is needed. In particular, an analytic change of coordinates does not make a pulled-back sharp indicator holomorphic. One either keeps its exact support, including all boundary terms when differentiating, or works on the original fixed source-free integration domain and varies only the source coefficients.

Where q has no regular chart, the pushforward measure or the original retained kernel remains meaningful. The differential formula in Section 6 is then stated through the kernel distribution, rather than inventing a smooth fiber Jacobian.

103.4.2 3.2 The original reference coordinate is not the old observable

Keep the full retained datum b fixed, including older variables that will be integrated later. The matched reference move has the form

$$\begin{aligned} D_0(b) &= \int \chi_{\mathrm{ref}}(C) r_b(CR(b))\, dC, \\ (\mathcal{R}_\alpha M)(C,b) &= \frac{\chi_{\mathrm{ref}}(C) r_b(CR(b))}{D_0(b)} \int q_0(\xi,b) M(\xi,b) \pi_\alpha(C,\xi,b)\, d\mu_b(\xi), \\ &\qquad \sum_\alpha \pi_\alpha(C,\xi,b) = 1. \end{aligned} \tag{XP13}$$

The relative tree measure is part of dC. The source note proves this formula from the fixed relaxed reference, the old reconstruction slot, the matching relative cutoff and tree measure, and the integrated output partition in R-II. All older omitted operators stay under their original integrals.

Consequently,

$$\sum_\alpha \int (\mathcal{R}_\alpha M)(C,b)\, dC = \int q_0(\xi,b) M(\xi,b)\, d\mu_b(\xi). \tag{XP14}$$

This holds for an arbitrary jointly gauge-invariant carrying function and hence for every coefficient of a finite source jet. No individual selected output history is asserted to have unit mass.

The old mark remains $M(\xi, b)$, not $M(CR(b), b)$. If one wants an observable of a physical coarse average of the old field, that average is itself filed in M before this move. XP14 does not identify the distribution of the reference coordinate with the physical coarse pushforward.

A gauge selected by a tree can also be applied to the old variables: the joint substitution has triangular Jacobian and unit Haar factors, and joint gauge invariance carries the whole mark. A colored subexpression is retained with its compensating transports or tensor indices until the full invariant object is formed.

The only inverse in XP13 is the fixed source-zero reference denominator from the original construction. Neither a sourced island mass nor the norm of a conditional sourced expectation is used.

103.5 4. Exact deterministic-tag extraction at every finite source degree

This is the algebraic result needed for the hybrid deterministic/proper-difference localization.

103.5.1 4.1 General factorization theorem

Let U denote the data held fixed in one conditional integration and let B denote **all variables integrated by that particular operation**. Write its unchanged source-zero weight as $K(U, dB)e^{V(U,B)}$. Suppose

$$A(\eta, U, B) \in \mathcal{J}_{[m]}, \qquad A(0, U, B) = 0.$$

If a zero-fluctuation value is defined, put

$$T(\eta, U) = A(\eta, U, 0), \qquad D(\eta, U, B) = A(\eta, U, B) - T(\eta, U). \tag{XP15}$$

Then $D(\eta, U, 0) = 0$, and the exact identity is

$$\int K(U, dB)e^{V(U,B)+A(\eta,U,B)} = e^{T(\eta,U)} \int K(U, dB)e^{V(U,B)+D(\eta,U,B)}. \tag{XP16}$$

It follows simply from commutativity of scalar source coefficients and independence of T from B. It is an unnormalized identity.

For an arbitrary multiplier $M(\eta, U, B)$ with $M(0, U, B) = 1$, first take $A = \log_{\mathcal{J}} M$. This finite logarithm introduces every mixed source coefficient exactly. Taking separate logarithms or deterministic parts of already truncated individual factors is unnecessary.

If T still depends on an old variable ξ, it factors only from a conditional integral that holds ξ fixed. It does not factor from a later island integral that integrates ξ. That later operation retains XP16 as an identity of its integrand.

103.5.2 4.2 The logarithm on a clean nonzero body

If the source-zero conditional scalar

$$Z_0(U) = \int K(U, dB)e^{V(U,B)}$$

is nonzero on the actual clean domain where a logarithm is used, the branch with source excess zero satisfies

$$\log \frac{Z(\eta, U)}{Z_0(U)} = T(\eta, U) + \log_{\mathcal{J}} \frac{\int K(U, dB)e^{V(U,B)+D(\eta,U,B)}}{Z_0(U)}. \tag{XP17}$$

This equality does not prove a uniform lower bound for $Z_0(U)$. On a still-live state, retain XP16 and do not introduce XP17.

For two labels let $A = \eta_1 A_1 + \eta_2 A_2 + \eta_1\eta_2 A_{12}$, with $A_I = T_I + D_I$. Direct extraction gives

$$\begin{aligned} [\eta_i] \log(Z/Z_0) &= T_i + \langle D_i \rangle_0, \\ [\eta_1\eta_2] \log(Z/Z_0) &= T_{12} + \langle D_{12} \rangle_0 + \langle D_1 D_2 \rangle_0 - \langle D_1 \rangle_0 \langle D_2 \rangle_0. \end{aligned} \tag{XP18}$$

The brackets here are only notation for the clean scalar ratios already present in XP17. The mixed contact coefficient T_{12} is not discarded. The product $D_1 D_2$ is integrated jointly.

103.5.3 4.3 Separate wall localizations are algebraically sufficient

Let $T_t(U)$ be a localized family with $T_{\mathbf{1}} = T$, independent of B, and let $D_s(U, B)$ satisfy

$$D_{\mathbf{1}} = D, \qquad D_s(U, 0) = 0. \tag{XP19}$$

They may have different wall parameters. Define the combined interpolating integrand using $T_t + D_s$, with whatever source-zero weakened kernel is actually being localized. It obeys the analogue of XP16 for every s, t for which it is defined. At all walls equal to one it is exactly the original marked integrand.

An exact multivariable fundamental-theorem-of-calculus expansion of this combined family therefore produces the same physical endpoint. There is no algebraic requirement that T_t equal the zero-fluctuation value of the independently chosen D_s family, or that two weakened minimizing maps commute at every wall value.

This statement licenses an exact representation, not its bounds. Each chosen family must have the required common domain, carrier support, and summable wall derivatives. All mixed derivatives of a product are retained; component factorization alone does not make arbitrary disconnected mixed derivatives vanish. Connected regrouping has to be performed before making the corresponding connected-support assertion.

103.5.4 4.4 Identification of a direct old tag

At the unweakened endpoint, if the old source slot is an actual reconstruction for a nested constraint with the same action and boundary problem, the variational identity PS2–3 identifies its zero-fluctuation value with a direct old critical readout. More precisely, HI1–HI3 proves the complex-domain identity $P_j P_k \simeq P_k P_j \simeq P_k$, where $P_j(U) = U_j(C_j U)$, on the actual common admitted branch. Target preservation proves one composition; the complex KKT equation and uniqueness prove the other. The exact constraint composition, rather than associativity of a single group mean, is what is used.

Consequently the source extension based at $P_{k+1}U$, as in HI7–HI8, has the direct deterministic value for every complex U in that common domain. This is stronger than identifying two values only at a real critical input. The common analytic domain and identical boundary/action problem remain necessary; the KKT identity does not supply their radius.

This identifies $T_{\mathbf{1}}$. It does not identify every independently weakened inner reconstruction with a true minimum. The original all-wall zero-B response identity used in IV is a different fact: it makes the **proper fluctuation difference** vanish coefficientwise at $B = 0$. It need not make the nonzero old determining data of a weakened inner reconstruction satisfy the unweakened variational problem.

Thus a finite vector of lower, current, contact, and remainder coefficients can be placed in the single polynomial A, split by XP15, and localized by XP19. Nothing is dropped in this operation.

103.6 5. The EOM sector is a differential module with contacts

103.6.1 5.1 Weighted divergence

For a smooth vector field X on the finite configuration space, let $\operatorname{div} X$ be its divergence for the reference Haar measure. Define

$$\mathsf{D}^{\dagger}_{\rho,X} F = -\rho^{-1} \operatorname{div}(\rho F X) = -XF - (\operatorname{div} X)F - (X \log \rho)F. \tag{XP20}$$

Then for every smooth G,

$$\int \rho\, \mathsf{D}^{\dagger}_{\rho,X} F\, G\, dU = \int \rho F\, XG\, dU. \tag{XP21}$$

For an invariant bond derivative X, Haar divergence is zero. With $\rho = e^{-S}$, XP20 is $(XS)F - XF$. The constants in the physically normalized current formula AI7–AI11 are obtained by multiplying this identity by the stated lattice and coupling factors.

The exact contact multiplication rule is

$$(\mathsf{D}^{\dagger}_{\rho,X} F)G = \mathsf{D}^{\dagger}_{\rho,X}(FG) + F\, XG. \tag{XP22}$$

Thus a completed EOM insertion has zero expectation against the constant test, but its multiples need not have zero expectation. It is not a multiplicative ideal that can simply be set to zero before a join.

An elementary compact example already shows the issue. On the circle with Haar probability and $X = \partial_\theta$, take $F = \sin\theta$. Then $\mathsf{D}^{\dagger} F = -\cos\theta$ has mean zero, while its product with $G = \cos\theta$ has mean $-1/2$. The contact on the right side of XP22 is $-\sin^2\theta$, with exactly the same mean. This is an obstruction to the proposed quotient operation, not an obstruction to carrying the full insertion.

103.6.2 5.2 Ordered products and source contacts

For vector fields $X_1, \dots, X_r$, repeated integration by parts gives

$$\int \rho\, \mathsf{D}^{\dagger}_{\rho,X_1} \cdots \mathsf{D}^{\dagger}_{\rho,X_r} F\, G = \int \rho F\, X_r \cdots X_1 G. \tag{XP23}$$

The order is reversed. Lie brackets and lower commutators cannot be discarded by treating these derivatives as commuting symbols.

For a nilpotent scalar source polynomial $A(\eta, U)$,

$$\begin{aligned} (\mathsf{D}^{\dagger}_{\rho,X} F)e^{A} &= \mathsf{D}^{\dagger}_{\rho,X}(Fe^{A}) + F(XA)e^{A}, \\ \int \rho(\mathsf{D}^{\dagger}_{\rho,X} F)e^{A} &= \int \rho F(XA)e^{A}. \end{aligned} \tag{XP24}$$

All terms are finite polynomials in the source labels. XP24 includes contacts with generated mixed coefficients A_I, not just with the original linear sources. Repeating it gives a finite ordered differential completion at any fixed degree. It does not assert that the resulting coefficients lie in a previously chosen finite mixing basis.

For a bond derivative X_b, $X_b O_i = 0$ if the original determining set of O_i omits b. These exact support zeros remain zero after transport. A later overlap of coarse representatives does not license creating or erasing a contact; the original anchors and determining bonds are retained until their exact coefficient has been formed.

The Wilson example AI14–AI15 illustrates why lower terms belong in the packet. The derivative of the Wilson force contains an explicit identity-plus-plaquette counterterm. Transporting the current square but discarding that counterterm changes the finite-cutoff observable.

103.8 7. Physical own-current ideals and field jets

103.8.1 7.1 A physical graph relation is not a quantum EOM identity

Let an admitted paired domain have independent coordinates (U, J), and let

$$\iota(U) = (U, J_{\mathrm{phys}}(U)), \qquad \mathcal{I}_{\mathrm{phys}} = \{F : \iota^* F = 0\}. \tag{XP34}$$

For the relevant source slot the physical graph may also include the actual nested critical reconstruction and its retained determining data.

A pullback Ψ^* preserves this vanishing ideal only if its input graph is mapped to the output graph:

$$\Psi^*\mathcal{I}_{\mathrm{phys,old}} \subseteq \mathcal{I}_{\mathrm{phys,new}} \quad \Longleftrightarrow \quad \Psi(\operatorname{graph} \iota_{\mathrm{new}}) \subseteq \operatorname{graph} \iota_{\mathrm{old}}, \tag{XP35}$$

where the equivalence uses the defining graph functions $J - J_{\mathrm{phys}}(U)$.

Equality on the graph also preserves only derivatives of the physical pullback. If X is a field direction, its tangent lift is

$$\widetilde{X} = X_U + (D_U J_{\mathrm{phys}}\, X) \cdot \partial_J, \qquad X(\iota^* F) = \iota^*(\widetilde{X} F). \tag{XP35a}$$

Iterated derivatives include derivatives of the coefficients of this lift. Independent ambient differentiation does not preserve the ideal: for the defining function $I = J - J_{\mathrm{phys}}(U)$, one has $\partial_J I = 1$. Thus physical equality of two extensions is not equality of their separate U- and J-jets.

Equality of two source extensions on that graph is sufficient inside a physical integral whose actual readout lies on it. This is PS8–9. It is not sufficient for equality on an arbitrary independent-current domain, or for every wall-weakened readout.

For example, the analytic graph $J = u^2$ is preserved at $s = 1$ by $\Psi_s(u, J) = (u, sJ)$, but the pullback of $J - u^2$ on the graph is $(s-1)u^2$. The unweakened endpoint identity does not imply the all-wall identity. Again this is a diagnostic of the algebraic implication, not a claim about a particular Wilson wall.

A classical critical relation is also not a pointwise relation under a fluctuation integral. The Gaussian saddle $y = 0$ does not imply that the integral of y^2 is zero. Conversely, the quantum weighted-divergence identity XP21 does not say its integrand vanishes on the classical graph. These are different quotients and must not be conflated.

103.8.2 7.2 Exact physical field-jet splitting

Let B_j be an admitted space of coefficient functions at scale j. Let $\mathcal{E}_j$ extract a specified finite collection of **physical** field jets after composing with the actual own-current map, and let $\mathcal{C}_j$ be a chosen completion with $\mathcal{E}_j\mathcal{C}_j = I$ on those jets. Then

$$F = \mathcal{C}_j\mathcal{E}_j F + (I - \mathcal{C}_j\mathcal{E}_j)F \tag{XP36}$$

is exact. The first part carries all lower/current/contact components selected by the extraction; the second part is retained as a function or joint kernel. Both are transported by XP6.

A completion may be added and subtracted before a join, but its inverse factor or compensating term is retained before taking absolute norms. Repeatedly bounding both sides of an exact cancellation at every scale need not give a summable estimate.

Neither the finite field-jet projection nor restriction to an independent-current Taylor polynomial generally commutes with integration. Field differentiation hits the kernel, reconstruction, Jacobian, and support. In a fixed smooth chart one has the ordinary Leibniz and chain-rule expansion of

$$\partial_y^\nu \int k(y,x) F(\Phi(y,x))\, d\lambda(x), \tag{XP37}$$

including every derivative of k and Φ; a moving cutoff contributes its boundary distribution. This is unlike source differentiation in XP6, for which the frozen kernel is source independent.

Thus exact finite-jet bookkeeping closes by retaining the complement and the full differential packet. It does not close by declaring EOM terms zero, by substituting an independent J for J_{phys}, or by asserting that integration preserves a finite field-polynomial algebra.

103.9 8. A quantitative interface that remains a single joint integral

The preceding statements are exact. The following elementary majorant records the norm input they actually require.

Suppose a fixed history has one positive majorant $G_\omega(x)\,d\lambda(x)$ for its **entire** joint source-zero integrand after the already-spent reserves have been removed. Suppose, before integration,

$$|A_I(x)| \le a_I(1+\mathscr{E}_\omega(x))^{d_I/2}, \qquad \int G_\omega e^{\theta\mathscr{E}_\omega}\,d\lambda \le B_\omega, \quad \theta>0. \tag{XP38}$$

Then XP4 gives

$$\left|[\eta_I]\int G_{\mathrm{actual}} e^{A}\,d\lambda\right| \le B_\omega \sum_{\pi\in\Pi(I)} \left(\prod_{B\in\pi} a_B\right)\left(1+\frac{\sum_{B\in\pi} d_B}{2\theta}\right)^{\frac12\sum_{B\in\pi} d_B}. \tag{XP39}$$

Indeed $(1+E)^q e^{-\theta E} \le (1+q/\theta)^q$ for $E\ge 0$, and the same energy reserve is used once for the total degree in each term.

There is only one factor B_ω in XP39. If two labels lie in the same rare live island, the formula does not invent two rarity factors. Distinct kernels can be multiplied only after a genuine conditional factorization with disjoint reserve ownership.

For fixed total source degree, XP39 is finite. A uniform hierarchy requires the constants a_I, d_I, θ, the carrier/age norms, and the terminal normalization to have controlled dependence on that degree. No such dependence follows from functoriality alone.

For the present D6 program, the proved canonical moment input PA10–13 and the clean unmarked inherited-vertex bound IV provide particular parts of XP38. They do not yet provide XP38 for every inherited marked coefficient after lower/current/contact joins. The full physical-core norm can include a factor g_j^{-d}, even when the fixed canonical external-ball norm is $O(1)$. Those two bounds cannot be interchanged in a recurrence.

Likewise, a deterministic old tag that has been factored by XP16 still needs its own admitted direct-readout norm. A proper difference with $D_s(U,0)=0$ can use a radial B-Schwarz estimate only when its actual common B-disc and the needed full coefficient norm have been established. The algebra does not enlarge that disc.

For sharp-history differential packets one may either cancel the complete partition boundary terms before taking norms, or prove bounds for the individual boundary currents in XP28–29. Pointwise domination of an arbitrary selected numerator does not by itself commute with connected vacuum cancellations.

103.9.1 8.1 The exact clean connected recurrence and its missing bounds

On an admitted clean branch with nonzero body, define the multilinear source-zero conditional cumulants $\kappa^{(0)}_{k,n}$ from that **same** integral. Expanding XP17 gives the exact identity

$$A'_{k,I} = T_{k,I} + \sum_{\pi\in\Pi(I)} \kappa^{(0)}_{k,|\pi|}(D_{k,B}: B\in\pi). \tag{XP40}$$

The one-block partition is the linear term $\kappa^{(0)}_{k,1}(D_{k,I})$. Every partition with at least two blocks is a proper source join. The formula includes already generated lower, current, and contact coefficients among the $D_{k,B}$.

For specified incoming and outgoing carrier norms, actual multilinear estimates

$$\left\|\kappa^{(0)}_{k,n}(F_1,\dots,F_n)\right\|_{\mathrm{out}} \le C_{k,n}\prod_{\nu=1}^{n}\|F_\nu\|_{\mathrm{in}} \tag{XP41}$$

would imply

$$\|A'_{k,I}-T_{k,I}\|_{\mathrm{out}} \le C_{k,1}\|D_{k,I}\|_{\mathrm{in}} + \sum_{\substack{\pi\in\Pi(I)\\ |\pi|\ge 2}} C_{k,|\pi|}\prod_{B\in\pi}\|D_{k,B}\|_{\mathrm{in}}. \tag{XP42}$$

Equations XP40–XP42 locate the remaining clean recurrence precisely. XP40 is proved algebraically; XP41 with the required grades, age weights, and completed supports is a quantitative input still to be established for the full marked class. A clean cumulant bound is not a bound for a still-live conditional numerator. At such a transition use XP9 and XP39, with the one joint cold kernel, until a genuine completed regular output has been formed.